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Computer Vision and Machine learning challenges in Cancer Research

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from knowledge generation to science-based innovation

INDEX

- Cancer Numbers and Facts
- CV & ML Challenges
- Cancer Research use cases & projects
    - Breast Cancer
    - Lung Cancer

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- Senior Research
  - INESC TEC Research Institute
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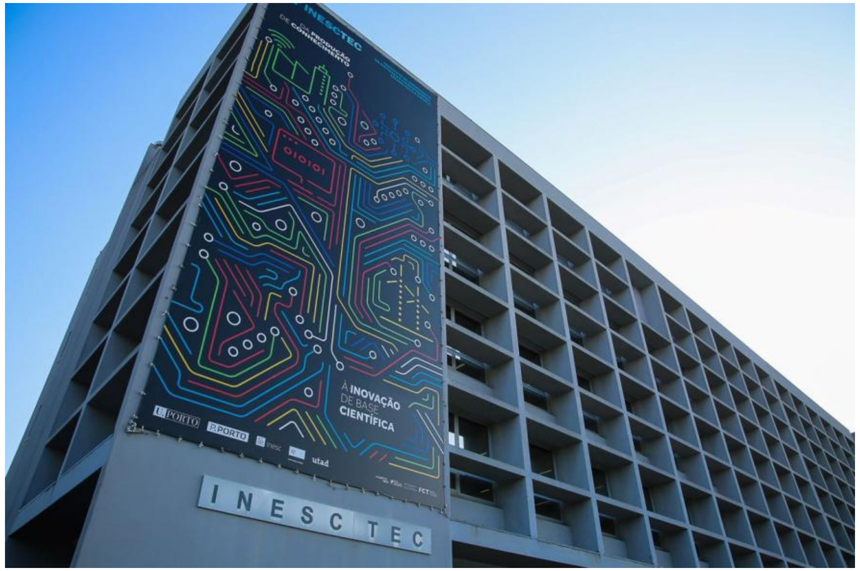
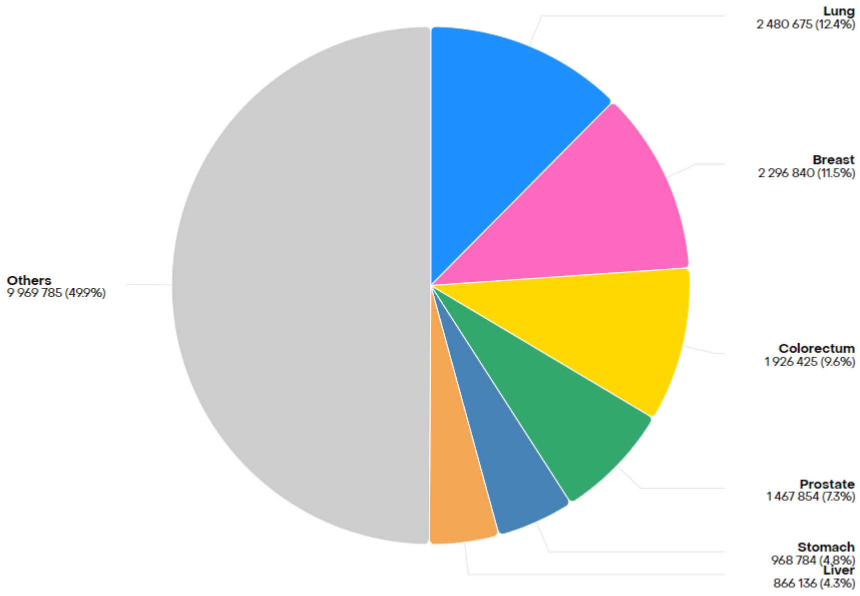


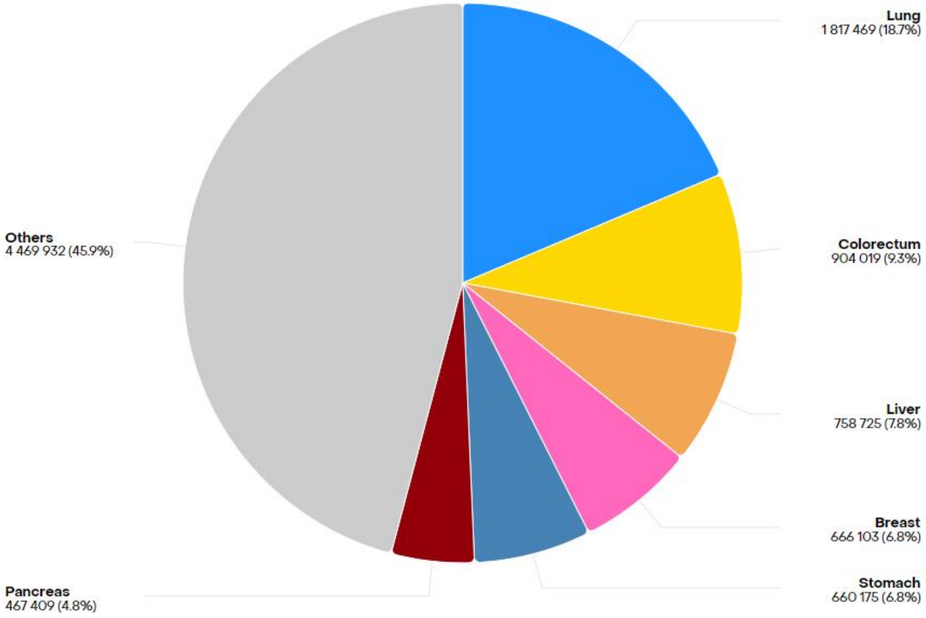
https://vcmi.inesctec.pt/
https://www.inescporto.pt/~hfpo/

CANCER NUMBERS AND FACTS

CANCER INCIDENCE AND MORTALITY



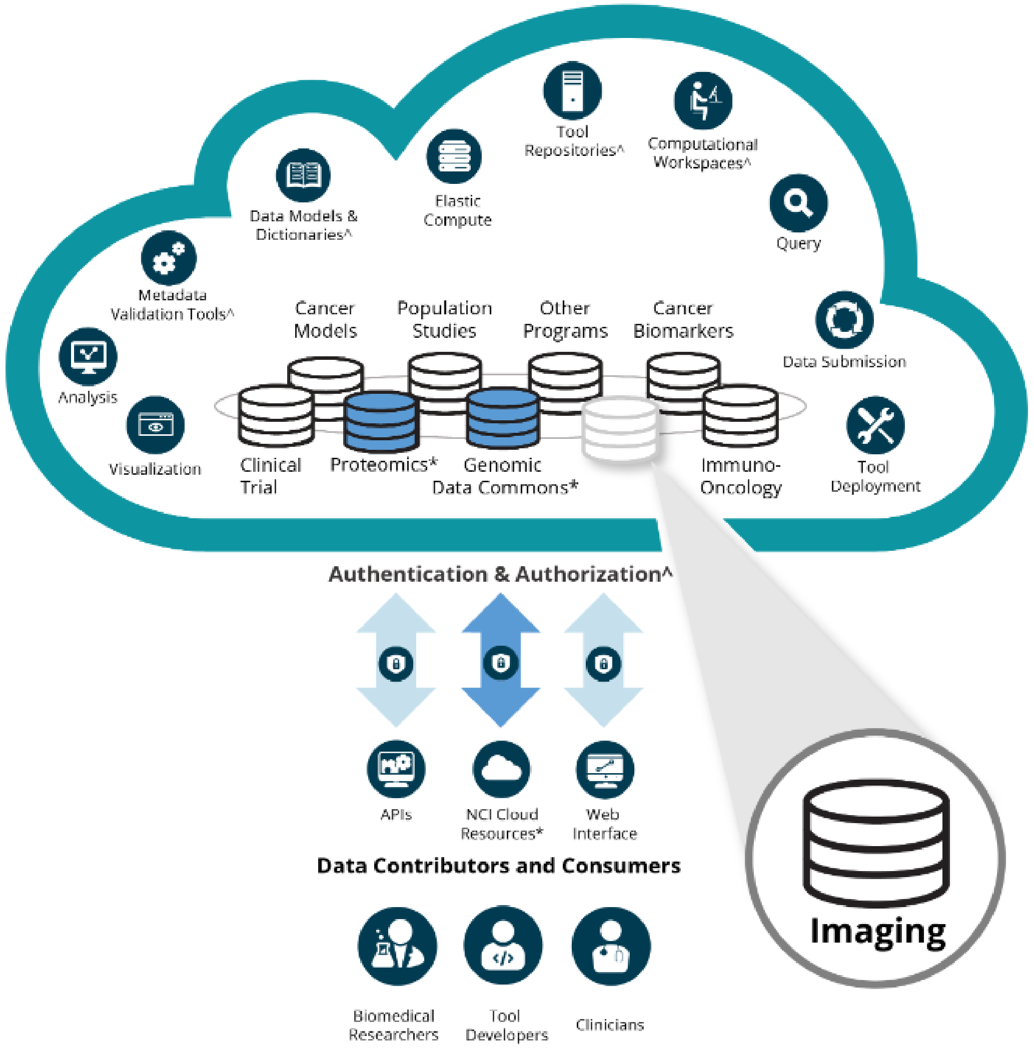
Total: 19,976,499
Cancer Incidence by Type in 2022



Total: 9,743,832
Cancer Mortality by Type in 2022

Data source: GLOBOCAN 2022
® International Agency for Research on Cancer 08.04.2025

CANCER RESEARCH DATA MANAGEMENT



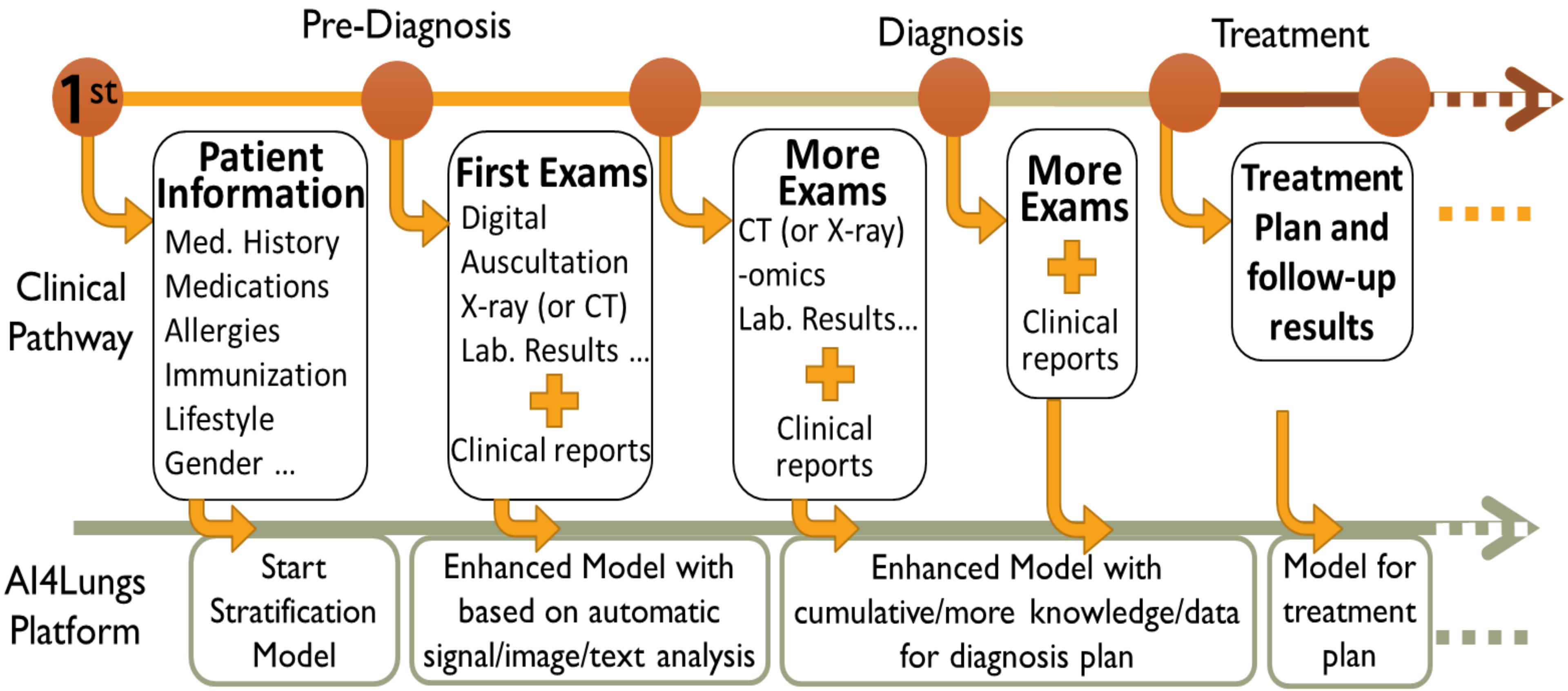
https://datascience.cancer.gov/news-events/blog/award-imaging-data-commons-bringing-multi-modal-imaging-data-cancer-research

P5 CANCER MEDICINE

- Predictive
- Personalized
- Preventive
- Participatory
- Psychology

L. Hood, et al. Predictive, personalized, preventive, participatory (P4) cancer medicine, Nature Reviews Clinical Oncology, 2011
Pravettoni G, Gorini A. A P5 cancer medicine approach: why personalized medicine cannot ignore psychology. J Eval Clin Pract. 2011 Aug;17(4):594-6. doi: 10.1111/j.1365-2753.2011.01709.x. Epub 2011 Jun 16. PMID: 21679280.

VISION: DIGITAL TWIN IN HEALTH

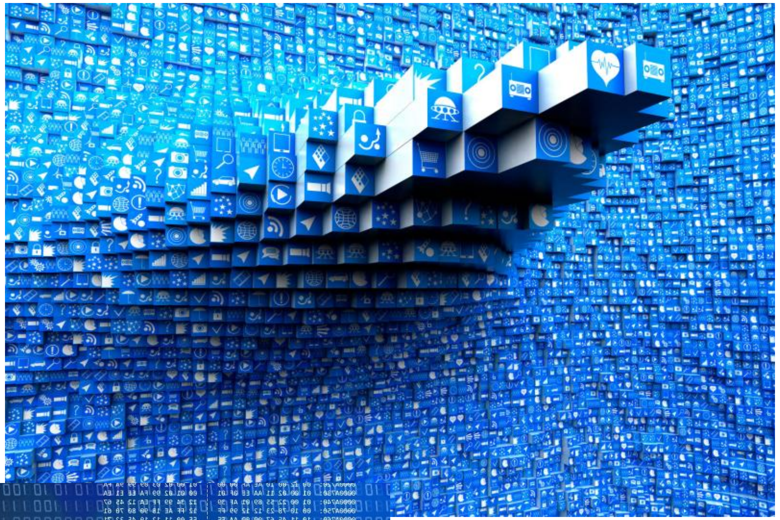


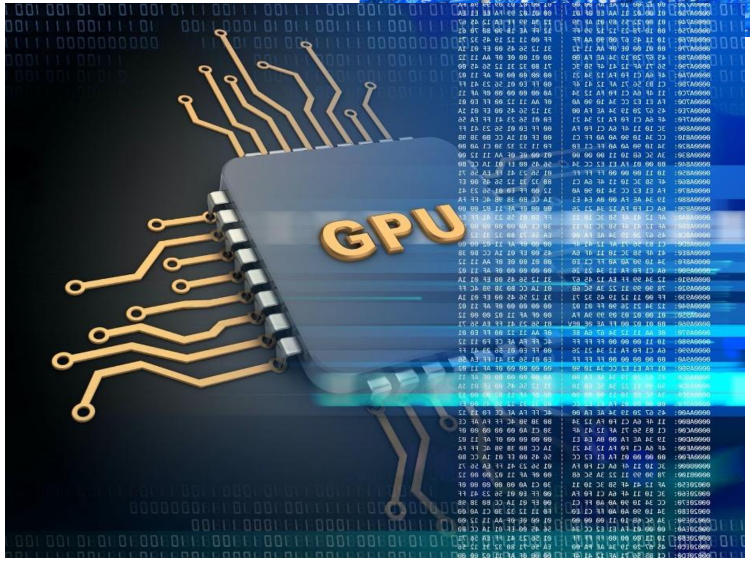
CV & ML CHALLENGES

HUGE DATABASES

- Size of medical image datasets (e.g. multimodalities, longitudinal studies, …), makes picture handling and representation calculations outdated

- Must be replaced by cutting edge parallelization techniques, by using supercomputers with GPU computation.



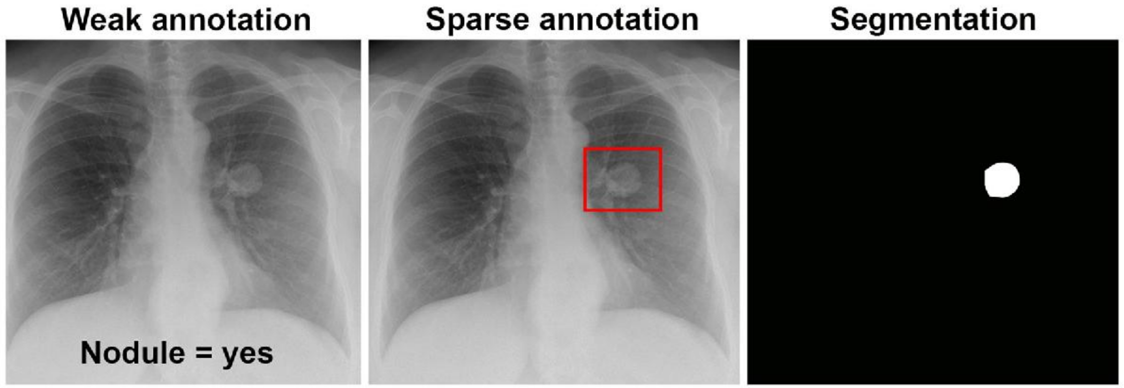


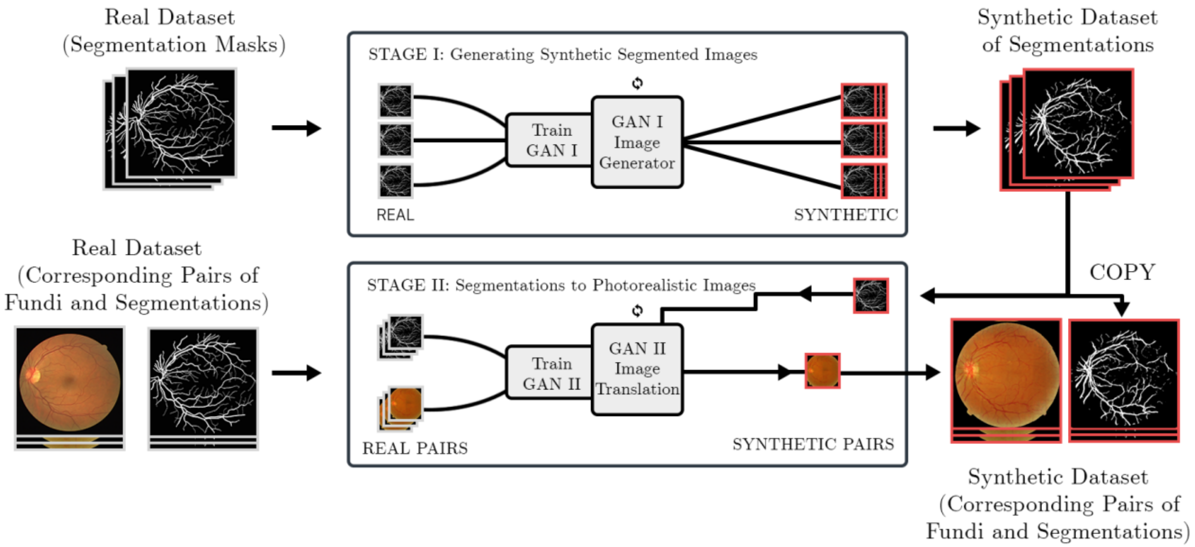
https://blog.equinix.com/blog/2016/08/01/speak-like-a-data-center-geek-big-data/

https://www.weka.io/blog/gpu-acceleration/

MISSING (WEAK) ANNOTATIONS

- Missing or weak labelling difficult the use of supervised methods
- Synthetization of artificial data could partially minimize this problem



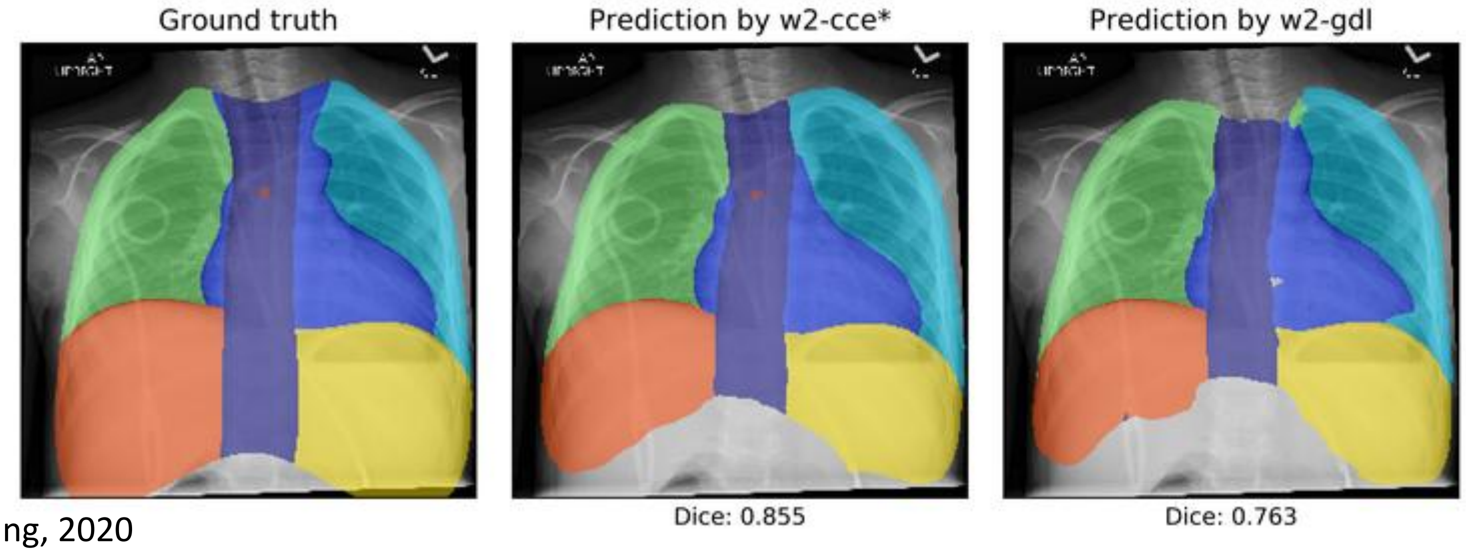


Chassagnon G, et al.. Deep learning: definition and perspectives for thoracic imaging. Eur Radiol. 2020 Apr;30(4):2021-2030. doi: 10.1007/s00330-019-06564-3.

arXiv: 1709.01872

IMAGE SEGMENTATION

- A solution for extracting anatomical structures for quantitative shape analysis or visualization.
- Should be fast, easy to use, robust to image artefacts, and, ideally, as automated as possible in clinical applications.
- Generalization for multiple datasets is the main issue
- The ultimate goal is to create structured visual representations from unstructured raw data.



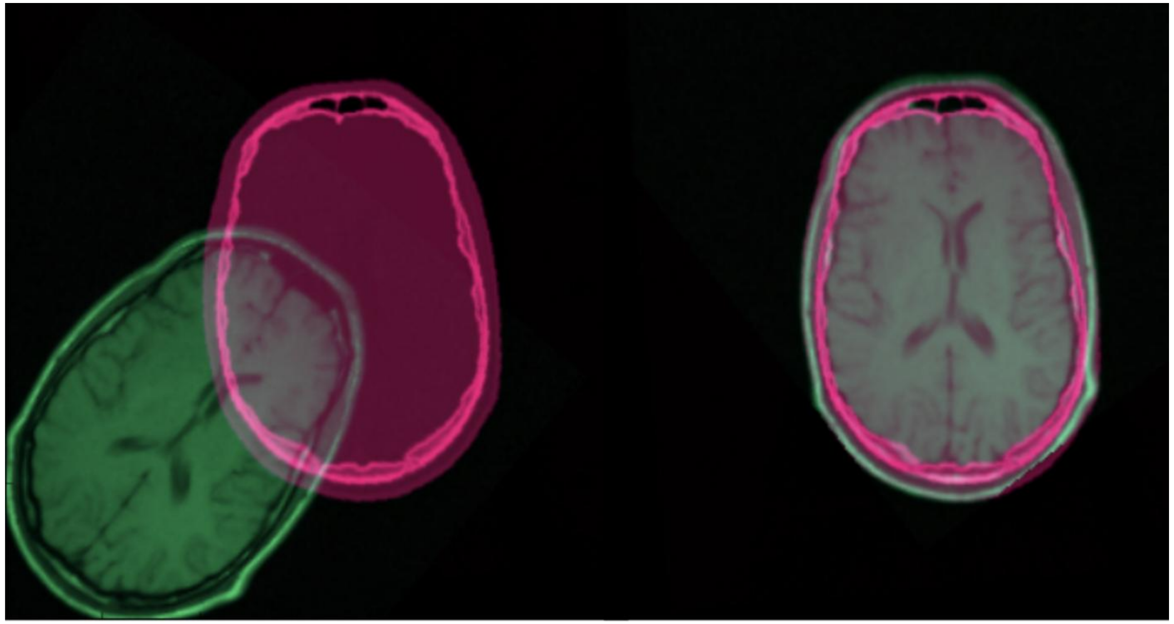
Gregory Holste, et al, SPIE Medical Imaging: Image processing, 2020

IMAGE REGISTRATION

- Aims to fuse images of the same region acquired from different modalities (e.g. MRI and CT)

- Or to interpolate corresponding images of a patient at different times or from different patients

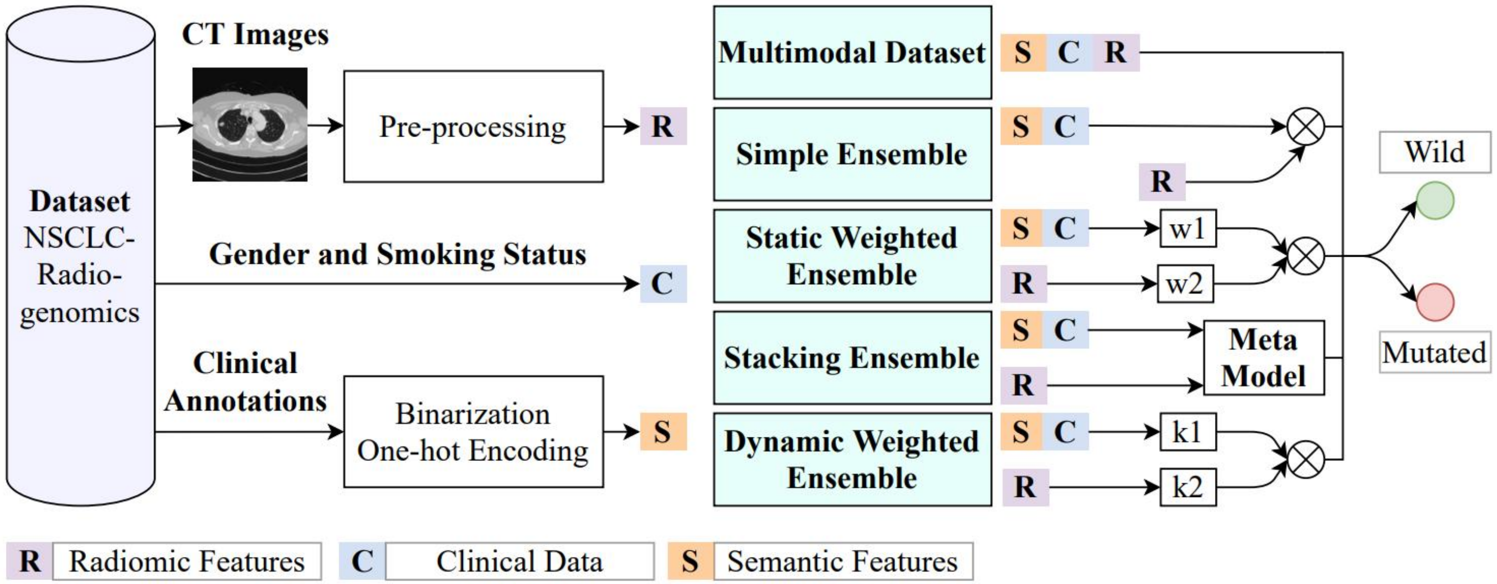
- Main issues are related with:
  - Different resolutions
  - Geometric deformations



Eman Kamoun, Image Registration: From SIFT to Deep Learning

ENSEMBLE-LEARNING

- Combine several base models to produce one optimal predictive model.
- Reduces the likelihood of giving weight to decisions made by a poor model.
- Regularization strategies and weighting factors based on domain knowledge
  - taking into account that physicians give different importance factors to each type of data in the moment of decision.

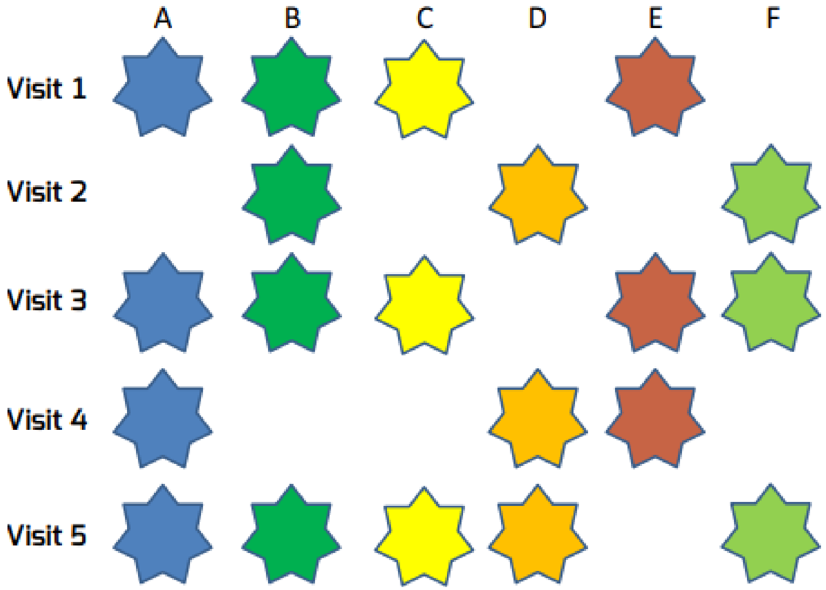


DATA SPARSITY, MISSING AND ANOMALOUS DATA

- Cause traditional predictive models to produce inconclusive results, hugely impacting performance.

- Deep learning techniques have shown promising results, but since they are post-processing methods, data consistency is inherently overlooked.

- Can be solved by a combination of entropy-based learning methods, with the objective to overcome the overfitting problem that emerges in the sparse and small sample data regimes by effectively learning common sparsity patterns that collectively form a compact and expressive representation of the source data.

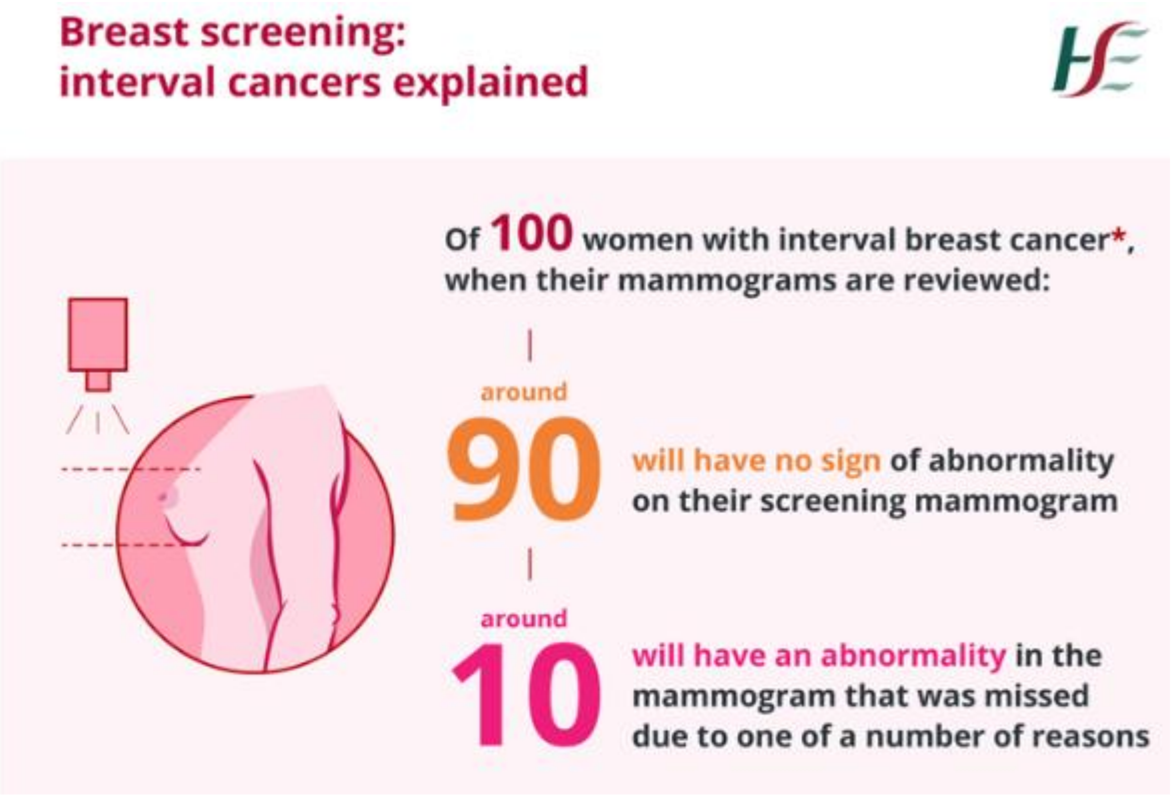


Thomas Schindler, Helen Bridge, The perils of the unknown: Missing data in clinical studies, Medical Writing, 2018

IMBALANCED DATA

- Most of the cancer problems are imbalanced

- Can be solved with
  - Under/oversampling methods
  - Cost-sensitive training

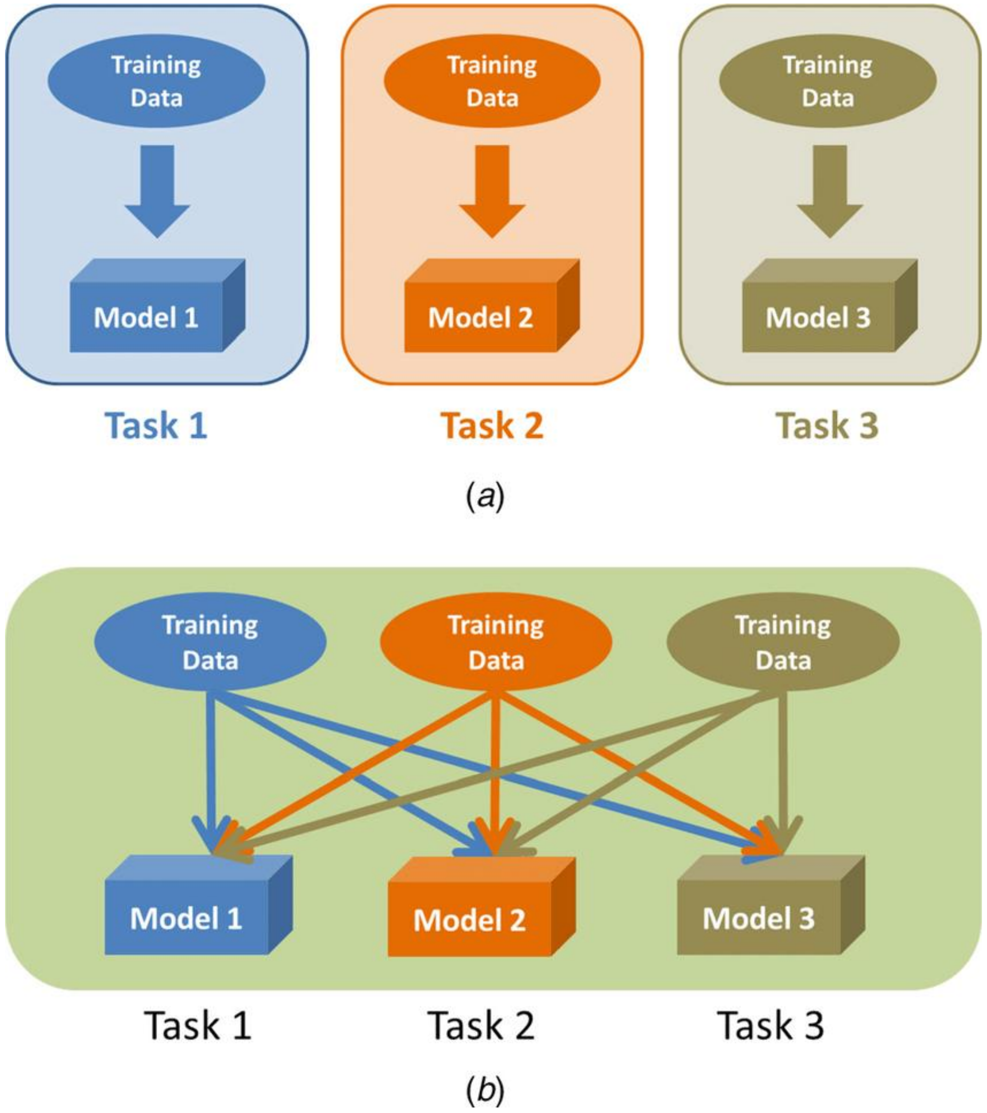


https://www2.hse.ie/services/breast-screening/benefits-and-limitations-of-breast-screening.html

MULTI-TASK LEARNING

- Multi-Task Learning (MTL) is a type of machine learning technique where a model is trained to perform multiple tasks simultaneously.

- In deep learning, MTL refers to training a neural network to perform multiple tasks by sharing some of the network's layers and parameters across tasks.



https://medium.com/@anilmatcha/summary-for-multitask-learning-by-rich-caruana-34427f7c7a87

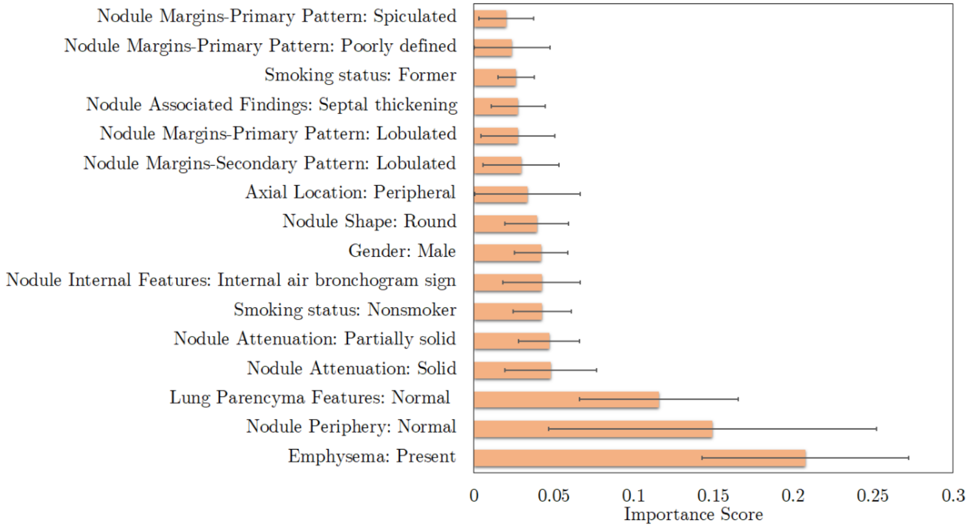
TRUSTWORTHY AND XAI

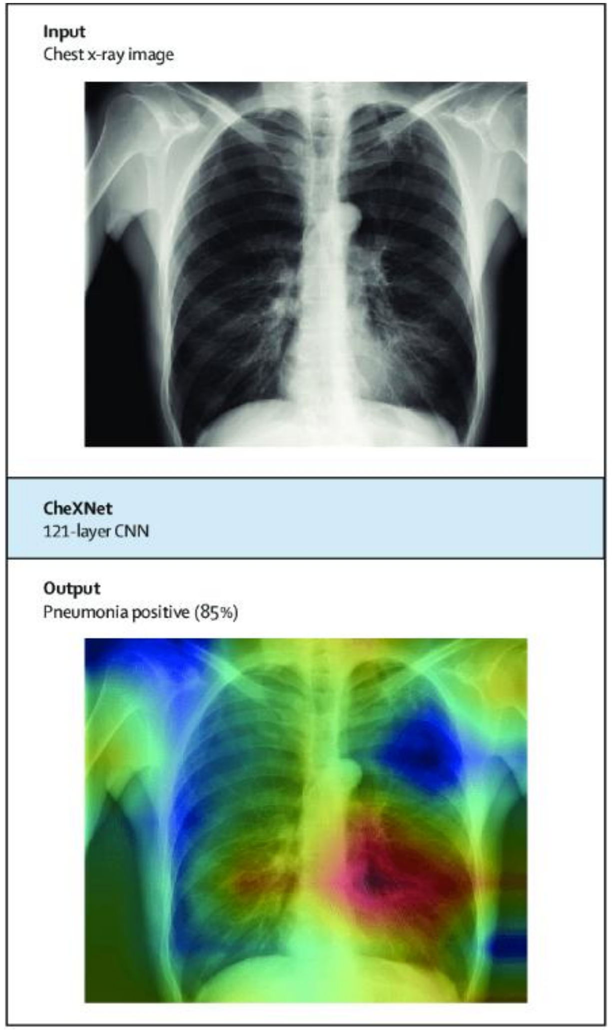
- Can significantly enhance the trust of medical professionals in future AI systems.

- Activation maps that visually highlight ‘hot spots’ e.g., in images, the most “important” regions from medical images.

- Detailing the most important features considered by the algorithm.

- Provide maximal transparency of methods.





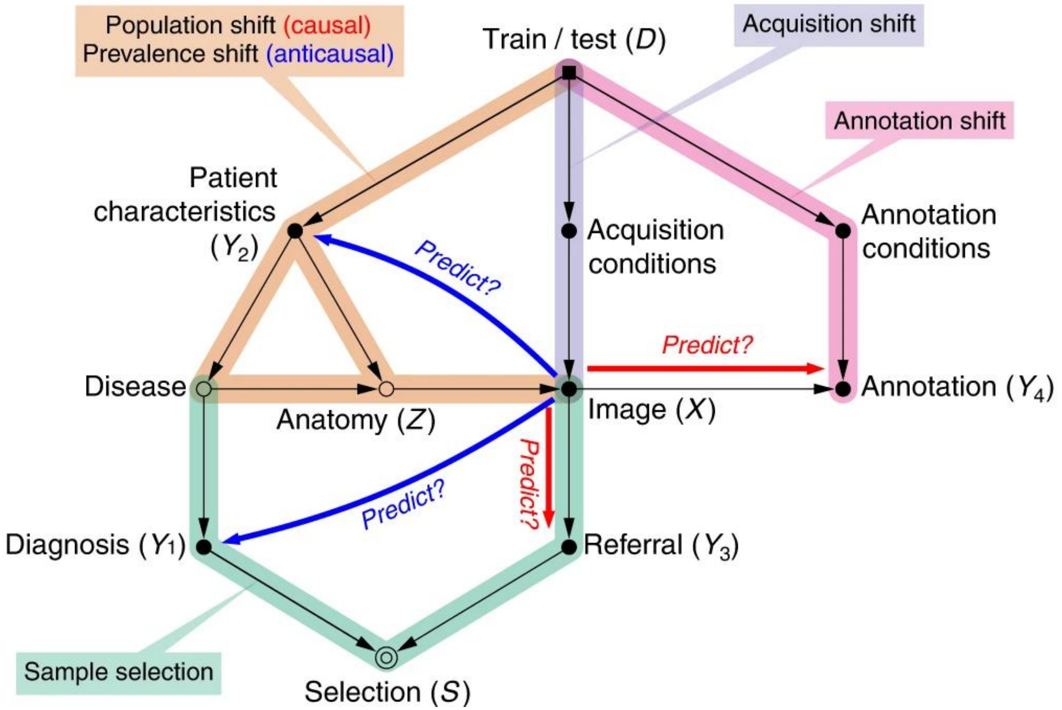
https://doi.org/10.1016/S2589-7500(21)00208-9

CAUSALITY

- Learning-based approaches have focused on predicting outcomes rather than understanding causality.

- Current deep learning models are good at finding data patterns, but can hardly explain how they connect or determine the cause-effect relationships.

- Causality can be use to help clinicians identify new leads and elements associated with disease development, which will lead to novel biological insights.

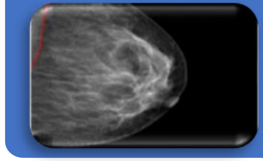
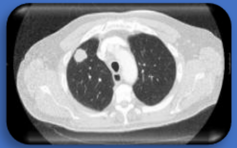
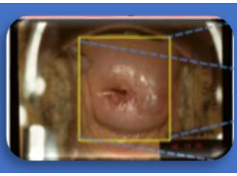
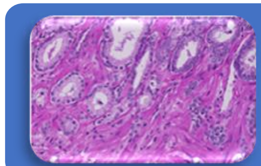
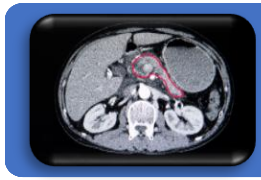
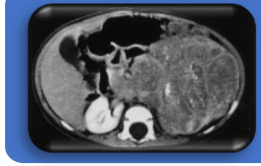


Castro, D.C., et al. Causality matters in medical imaging. *Nat Commun* **11,** 3673 (2020). https://doi.org/10.1038/s41467-020-17478-w

USE CASES

CANCER RESEARCH

IPOPORTO
SÃO JOÃO HOSPITAL
30 ANOS A DESCOBRIR O CANCRO ipatimup
CENTRO HOSPITALAR VILA NOVA DE GAIA|ESPINHO
IMP DIAGNOSTICS MOLECULAR & ANATOMIC PATHOLOGY
HEIDELBERG UNIVERSITY HOSPITAL
SHEBA Tel HaShomer | GLOBAL Patient Services
LUMC
Vall d'Hebron Hospital
I.R.C.C.S. Ospedale San Raffaele
Gruppo San Donato
Medical University of Gdansk
Amsterdam UMC University Medical Centers
Oslo University Hospital
NIPH Norwegian Institute of Public Health
First
RABIN MEDICAL CENTER BEILINSON • HASHARON
Mor Research Applications LTD
NHS Royal Free London
bmd
IT PEOPLE INNOVATION
BYME health care

	Computer Vision	Machine Learning
 Breast Cancer		
 Lung Cancer		
 Cervical Cancer		
 Prostate & Colorectal Cancer		
 Pancreatic Cancer		
 Neuroblastoma Cancer		

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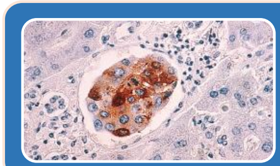
BREAST CANCER

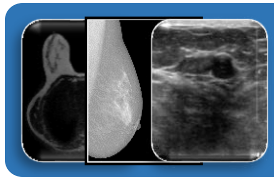
BREAST CANCER RESEARCH LINE

Computer Vision

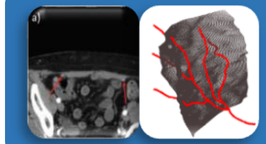
Machine Learning



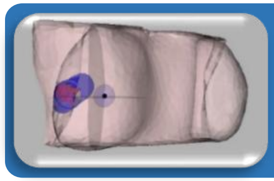
Computational Pathology



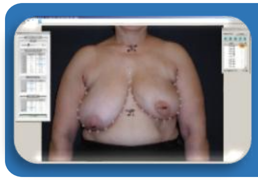
Radiological Image Analysis



Surgery planning (Reconstruction Surgery)



Surgery planning (Conservative Treatment)



Surgery Evaluation

Rehabilitation

BREAST CANCER

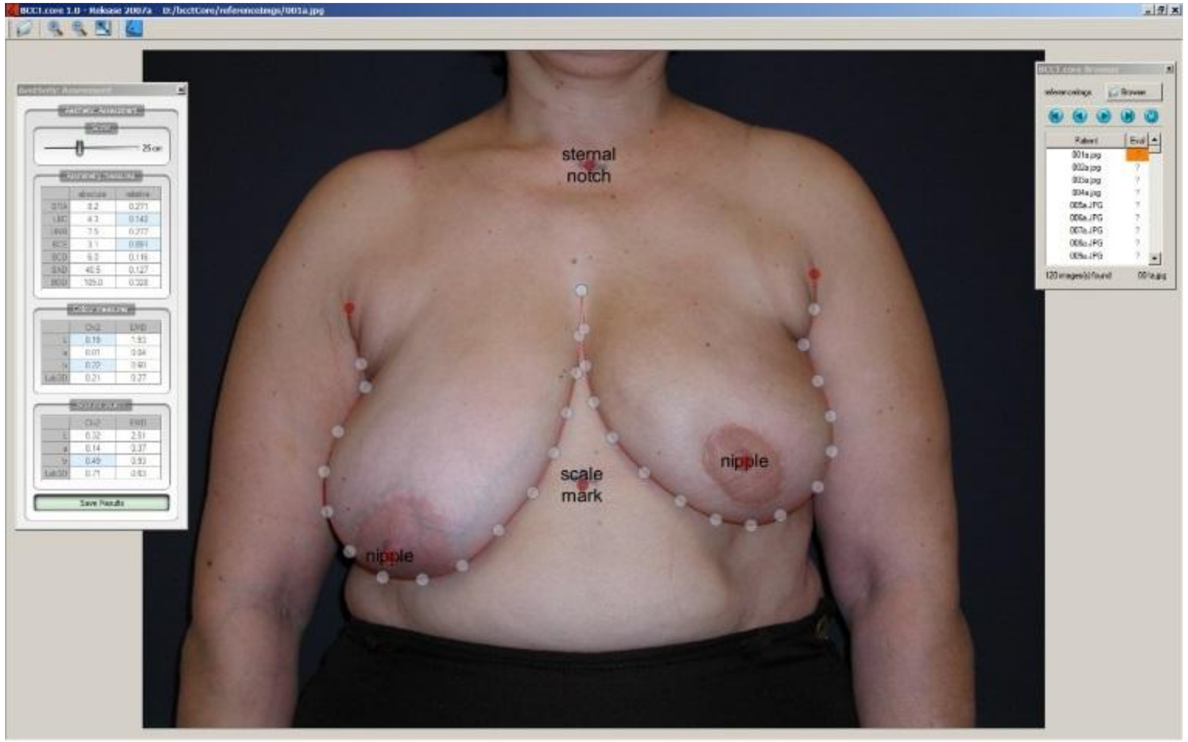
SURGETY PLANNING

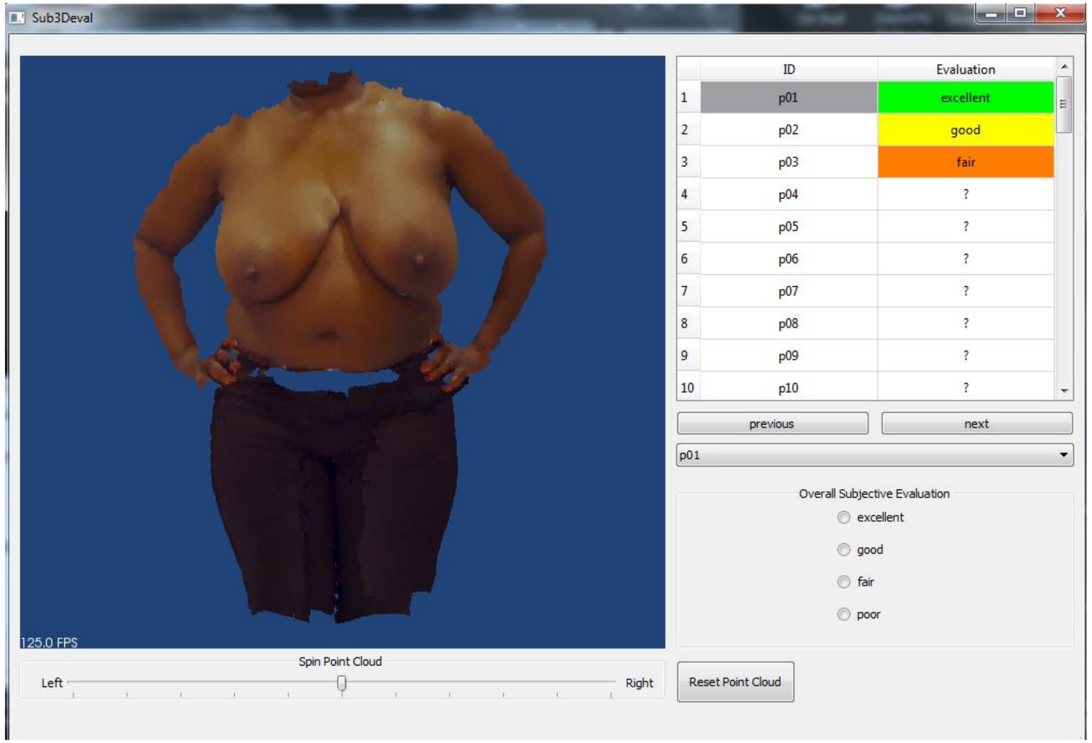
CONSERVATIVE TREATMENT

Motivation

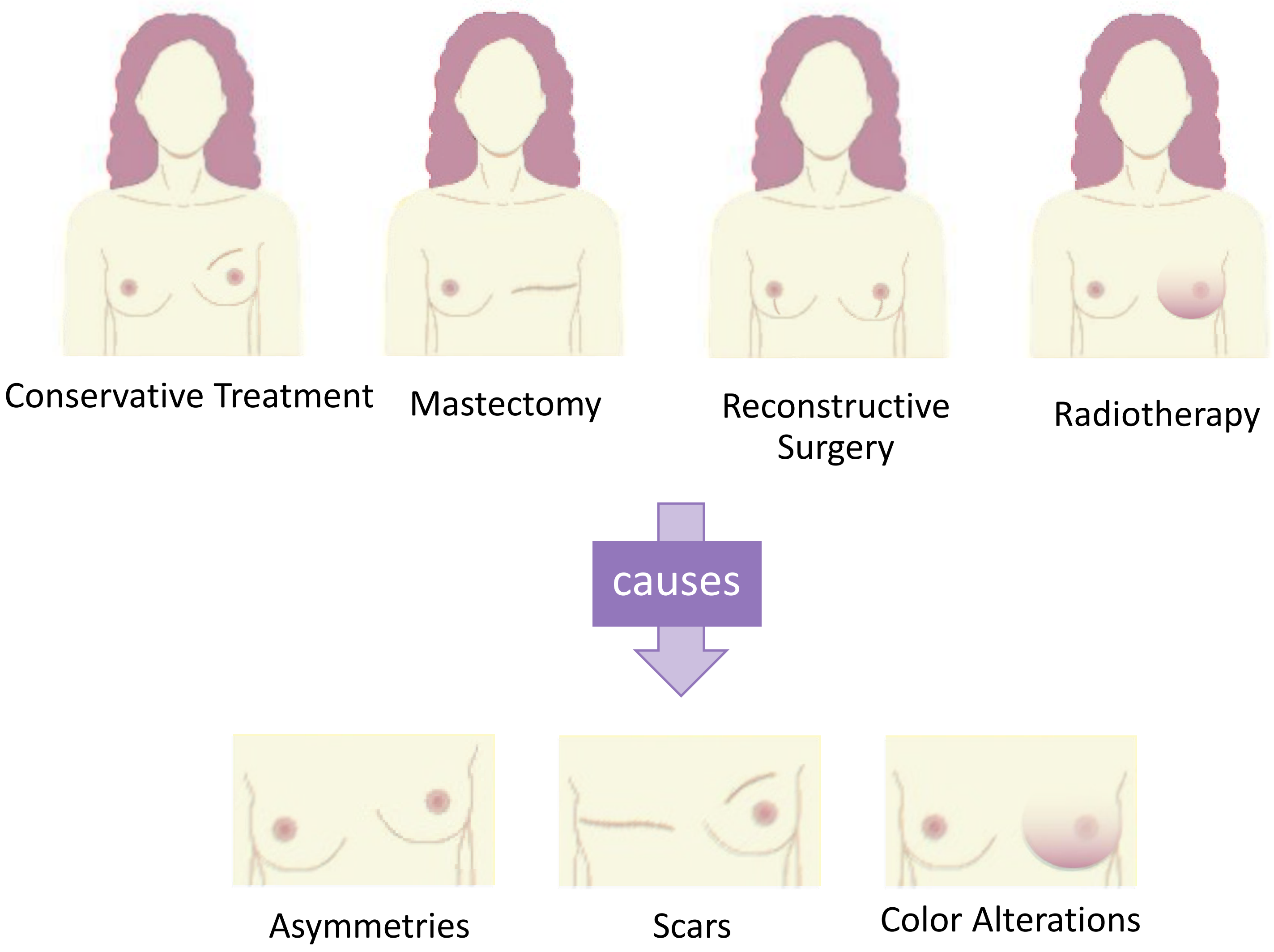
https://www.youtube.com/watch?v=O7-6BZDjcqQ

THE BEGINNING





THE PROBLEM

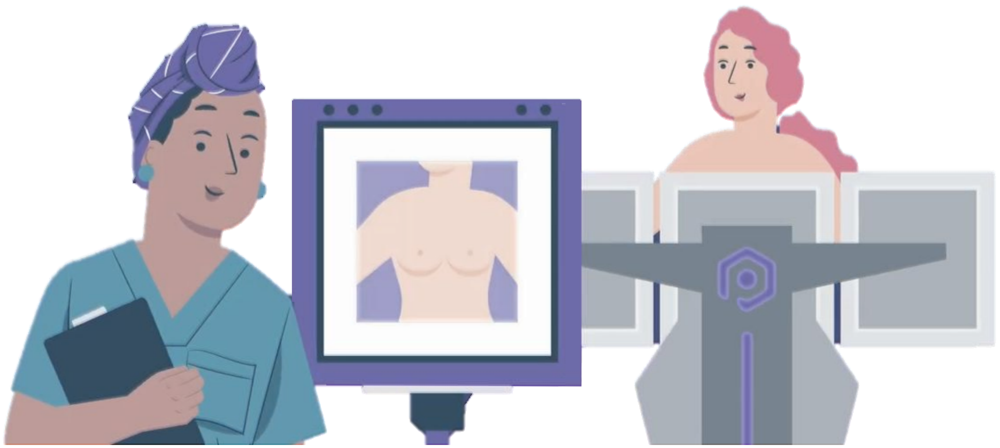

Conservative Treatment
Mastectomy
Reconstructive Surgery
Radiotherapy
causes
Asymmetries
Scars
Color Alterations

THE PROBLEM

Patients often experience:

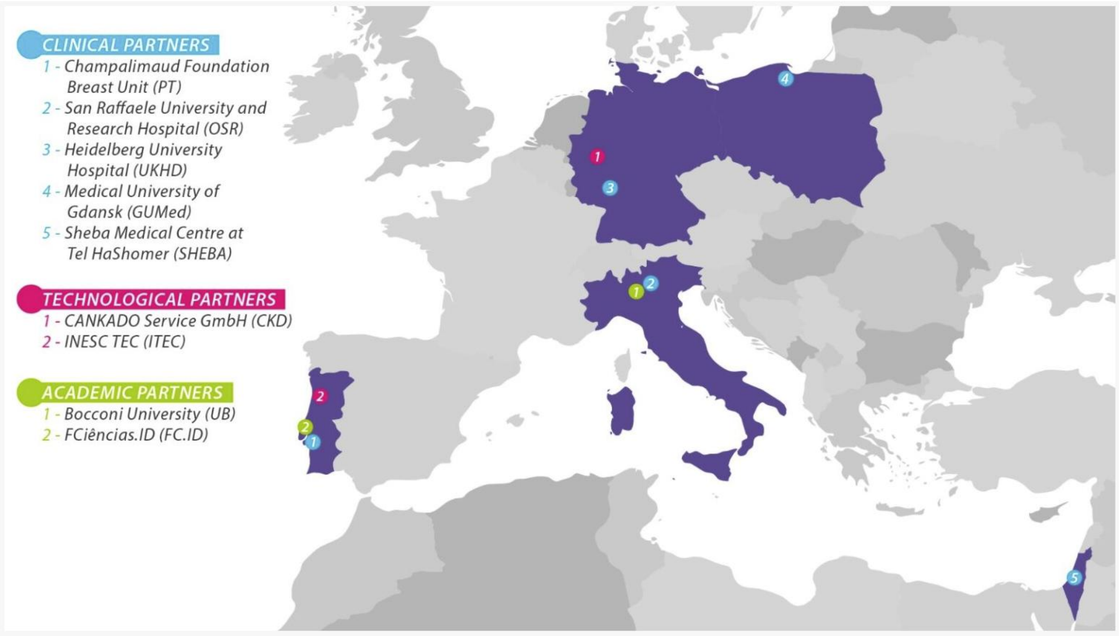
- Difficulty in choosing the treatment plan
- Disappointment on the outcomes of treatment
- Lower self-esteem after treatment

EU CINDERELLA PROJECT



Objectives

- Use the existing BRELO-AI SYSTEM
- Establish a routine digital approach
- Develop a data-friendly and AI-friendly culture

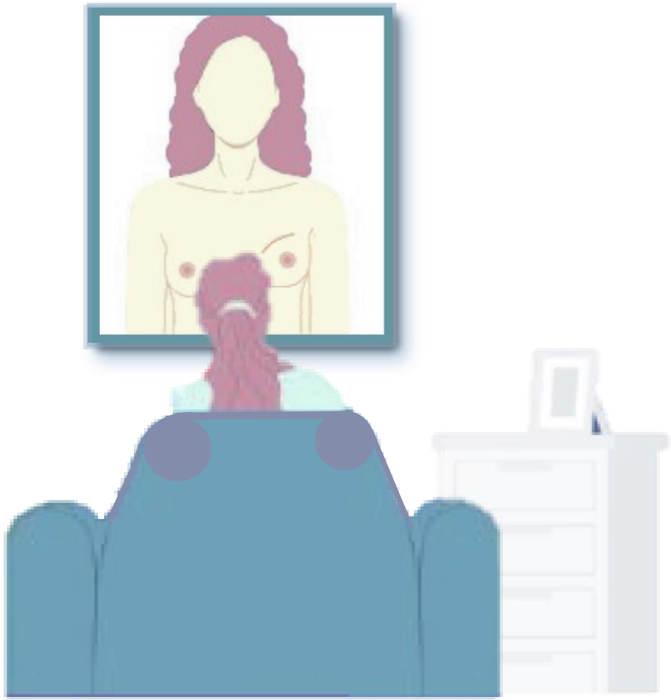


Web-based app for decision support

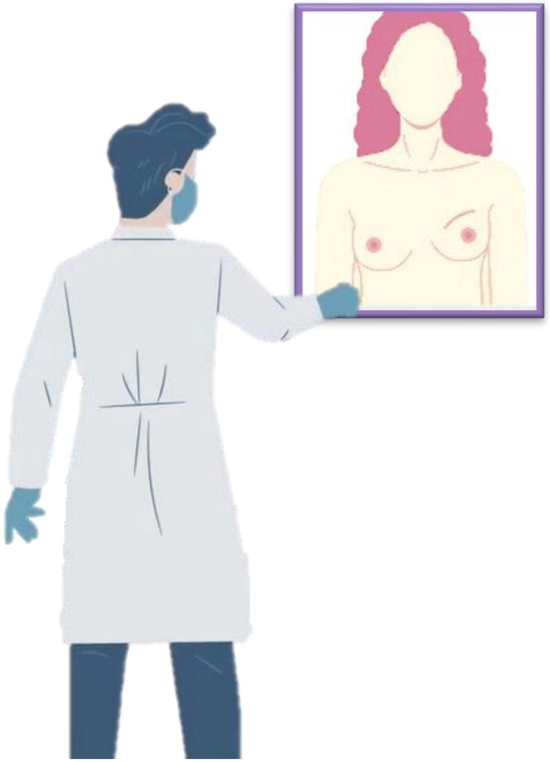
- Patient’s picture is uploaded to the app
- App shows how the patient may look like after treatment

EU CINDERELLA PROJECT

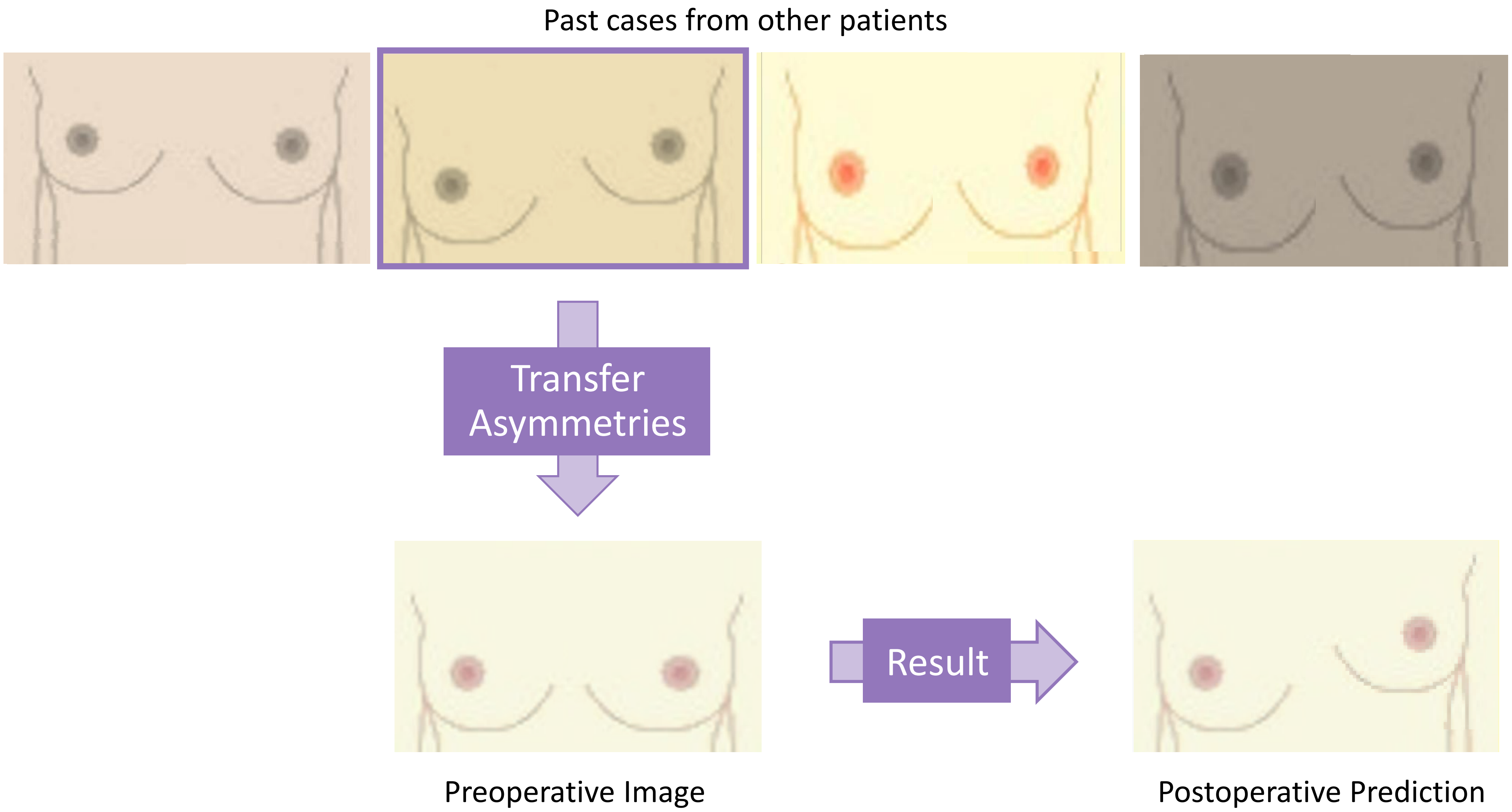
Predicting the Aesthetic Outcome



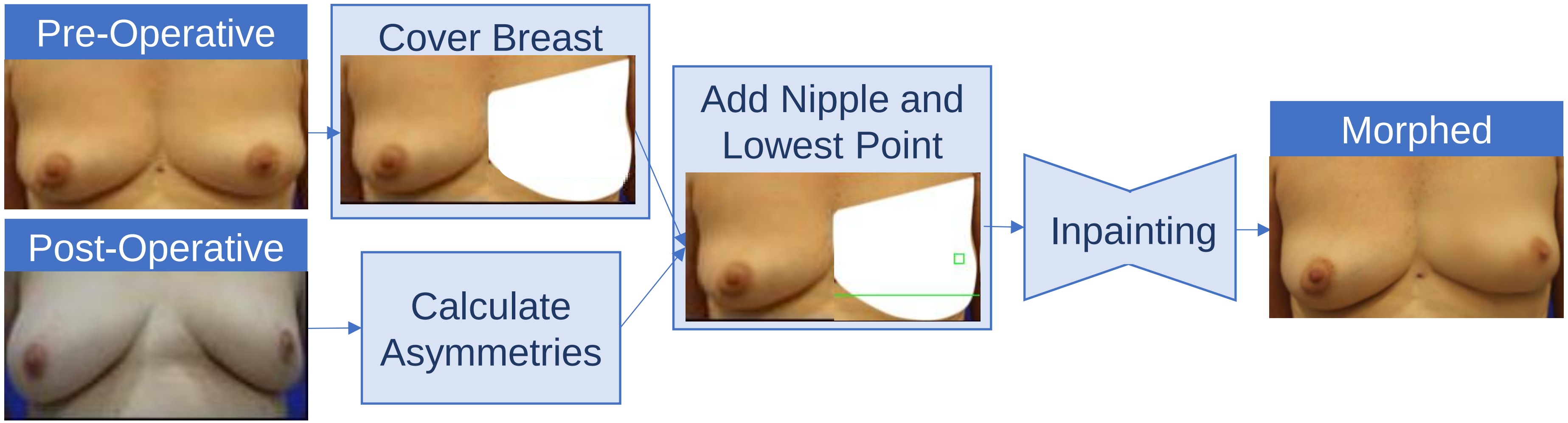
Evaluating the Aesthetic Outcome



PREDICTION OF THE OUTCOME OF BREAST CANCER TREATMENT



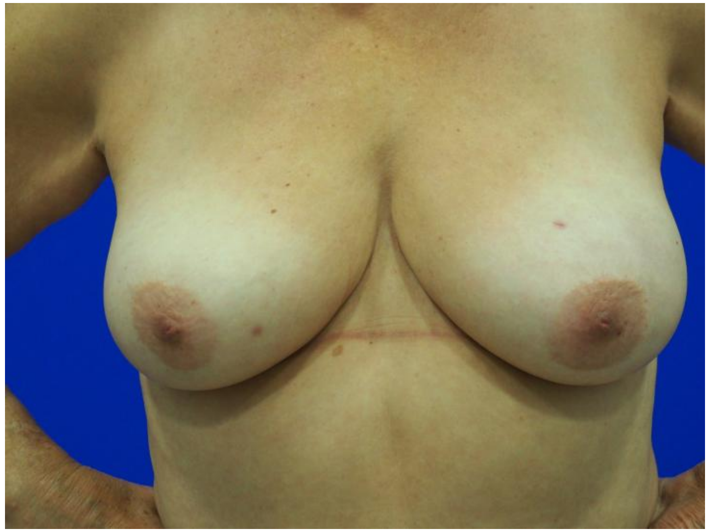
INPAINTING MODEL



INPAINTING MODEL RESULTS

Post-Operative Past Case

Pre-Operative Image



Post-Operative Prediction

EU CINDERELLA PROJECT – ALL INTEGRATED

Brelo-AI Platform

Video:

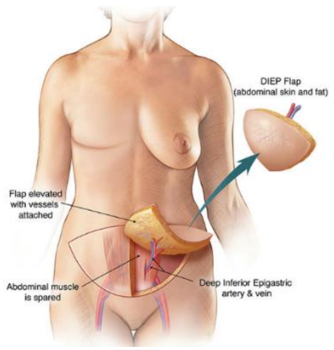
BREAST CANCER

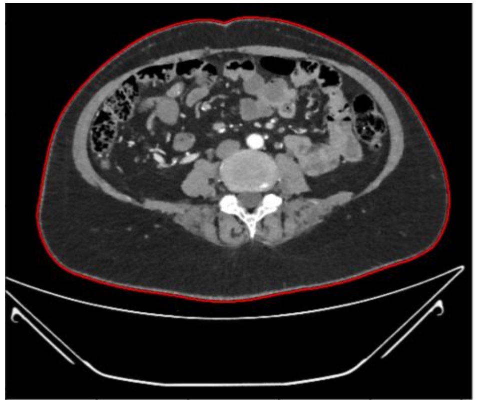
SURGETY PLANNING

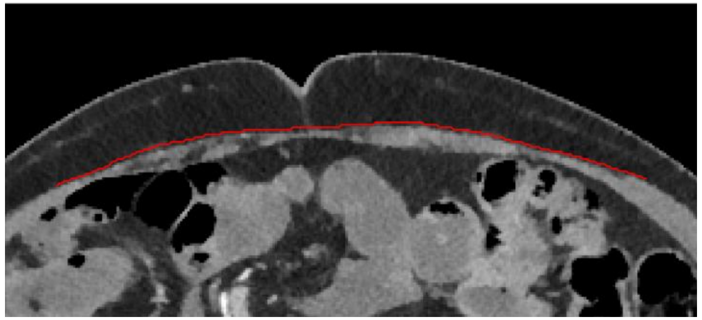
Reconstruction Surgery

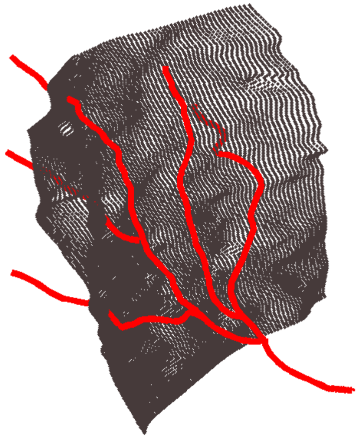
VISION

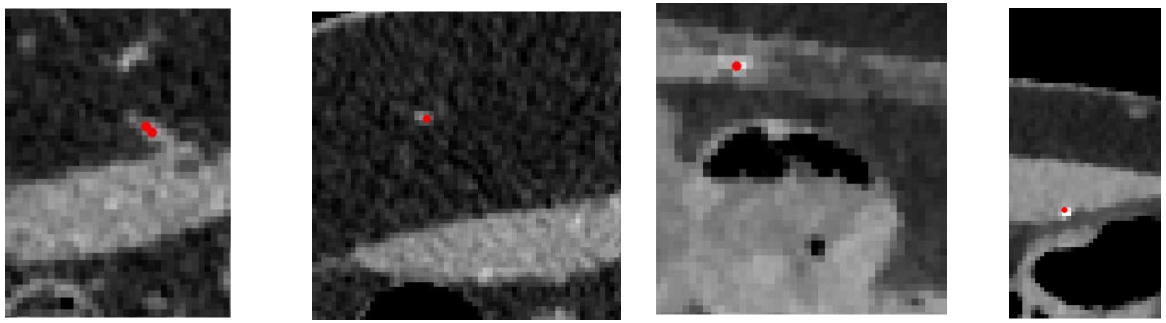
SURGERY PLANNING – RECONSTRUCTION SURGERY



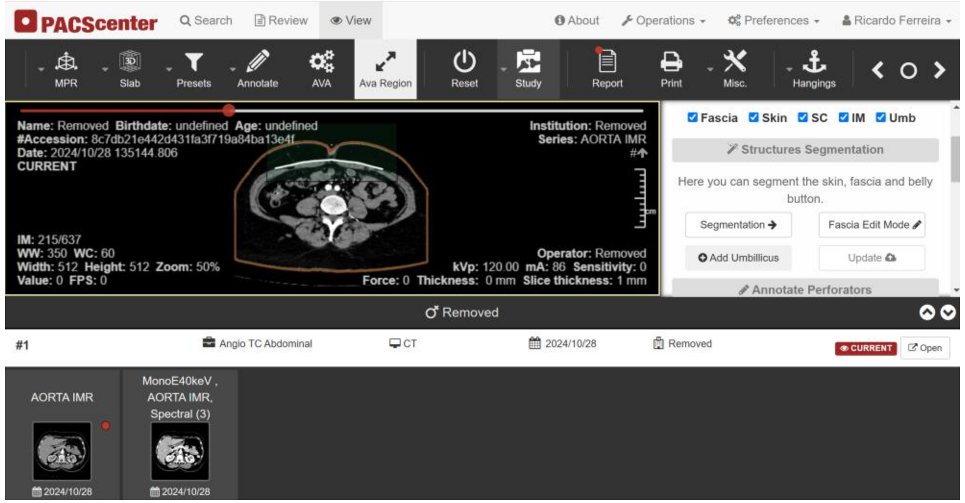






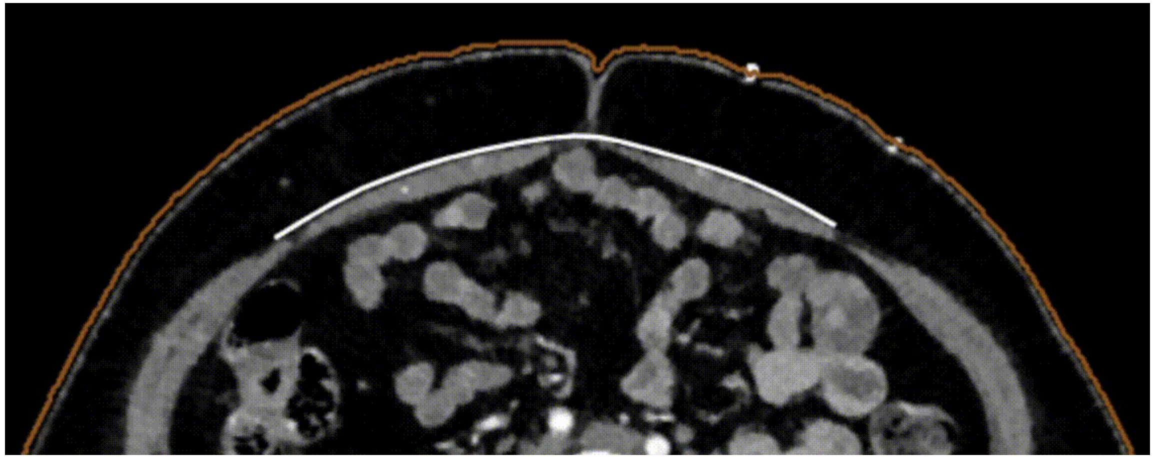


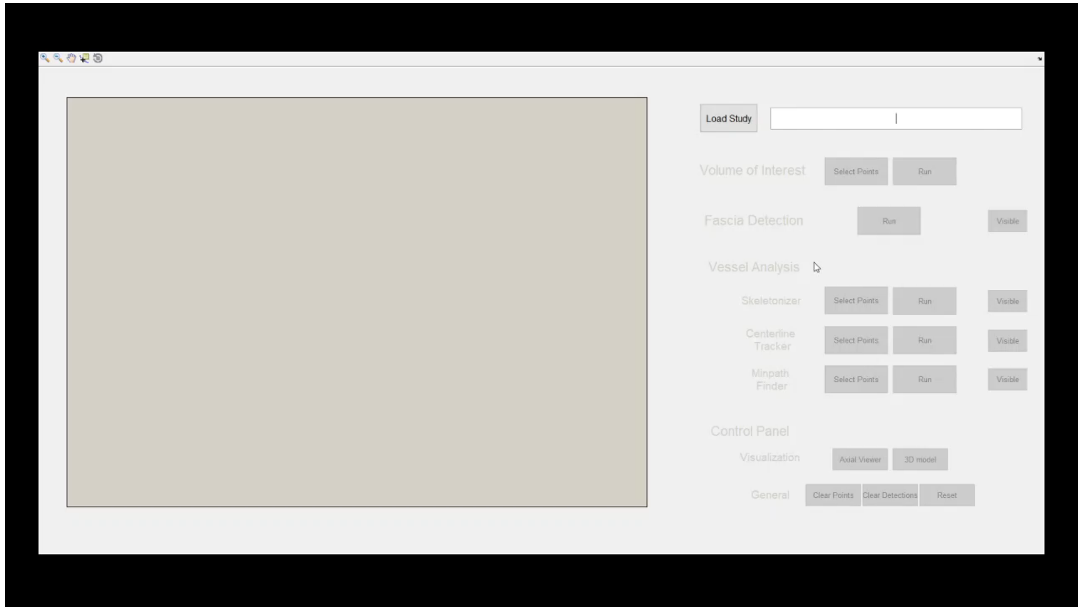
SURGERY PLANNING – RECONSTRUCTION SURGERY



Report AVA_080

Perforator	Caliber	Start Coordinates (mm)	Coordinates Fascia Intersection (mm)	Intramuscular Path Length (mm)
P1	2.62	27 right 28 up	17 right 10 up	56.67
P2	2.55	20 right 3 down	11 right 7 down	12.17





SURGERY PLANNING – 3D VISUALISATION

CLINICIAN-PATIENT COMMUNICATION

MEDICAL TEAM COMMUNICATION

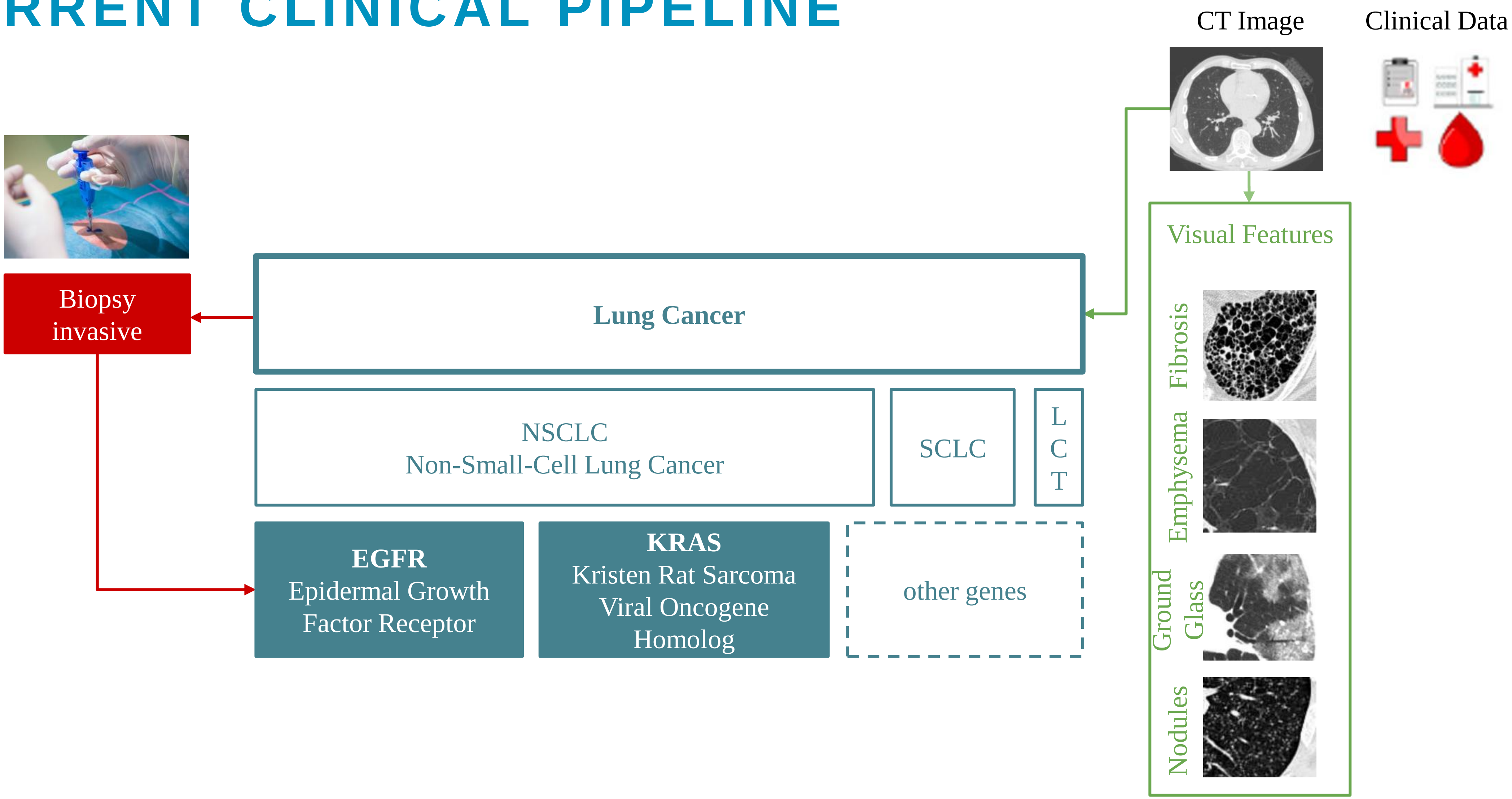
TRAINING

Neadvance
Intelligent Vision Systems

INESCTEC

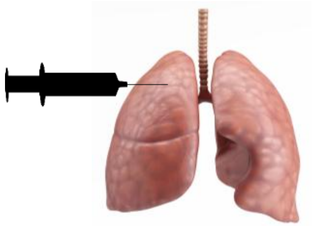
LUNG CANCER

CURRENT CLINICAL PIPELINE



TREATMENT DECISION

Biopsy



Genotype

Gene mutation status

<u>Treatment plan:</u>

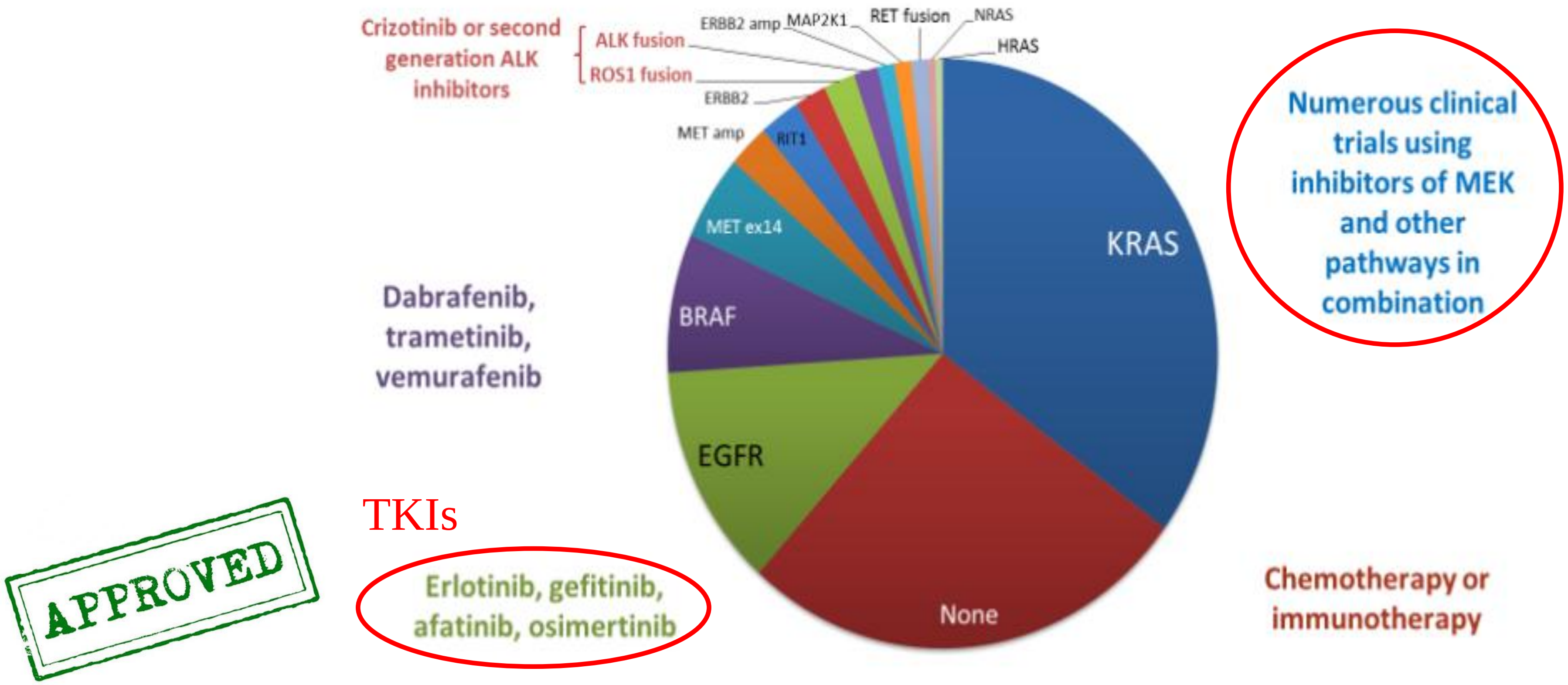
1. Chemotherapy/ Radiotherapy;

Traditional Therapies

2. Tyrosine kinase inhibitors (TKIs);
3. Immunotherapy.

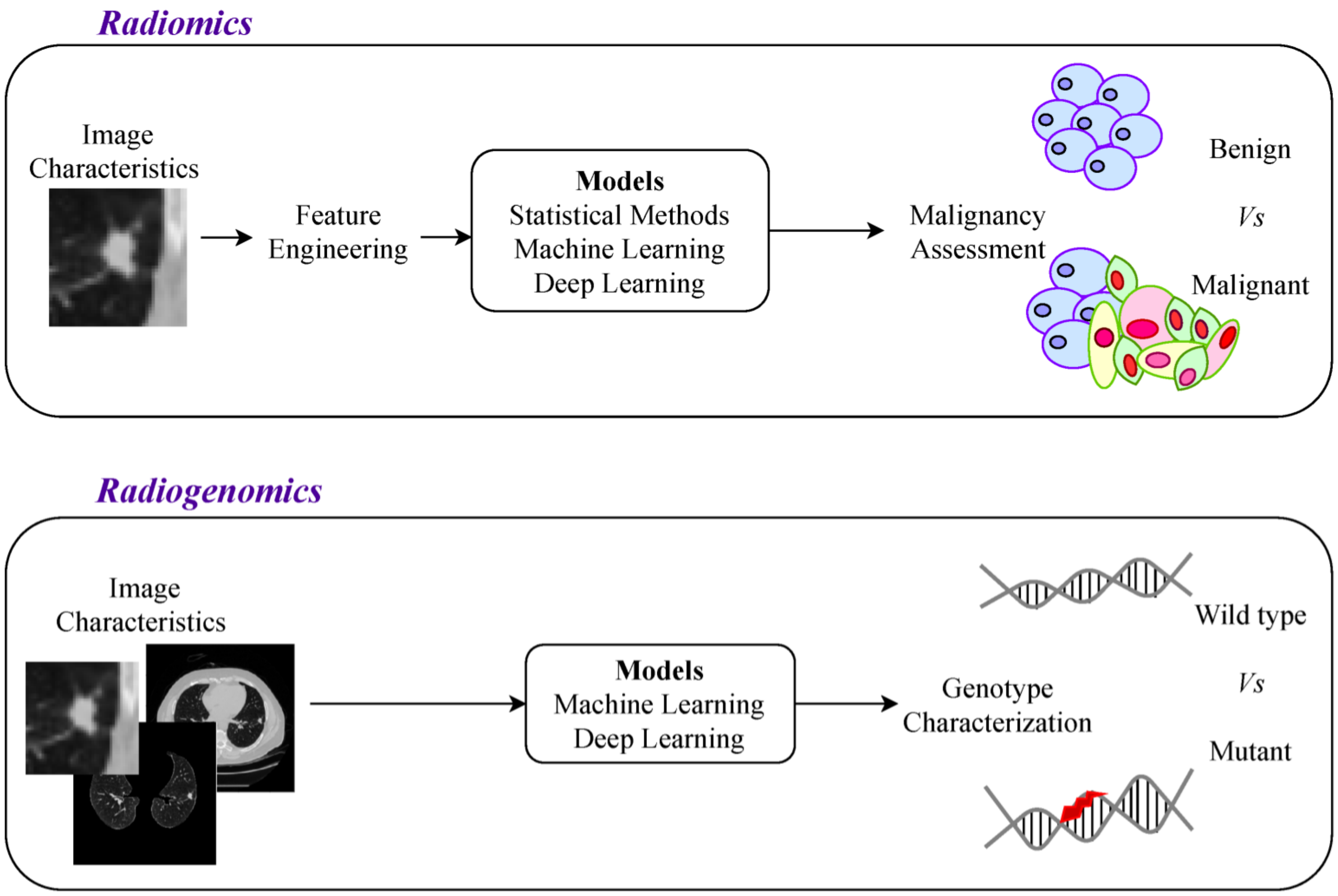
Target Therapies

MOLECULAR ANALYSIS AND TREATMENTS

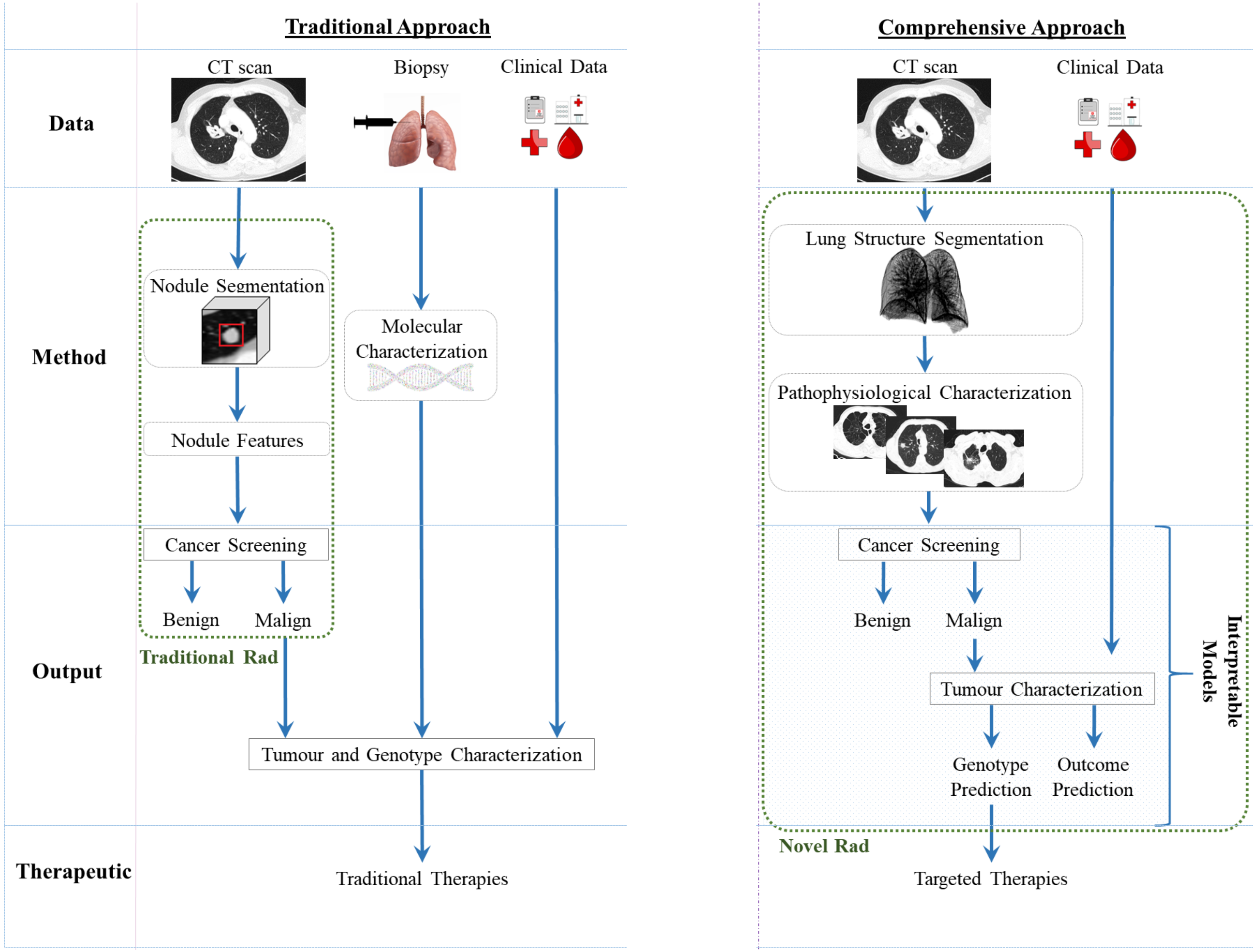


From: “Translation of Molecular Genetics in Lung Cancer into Treatment Decision” in https://cancergrace.org/post/translation-molecular-genetics-lung-cancer-treatment-decision

RADIOMICS VS RADIOGENOMICS



TRADITIONAL CADS VS COMPREHENSIVE CADS



Journal of *Personalized Medicine*

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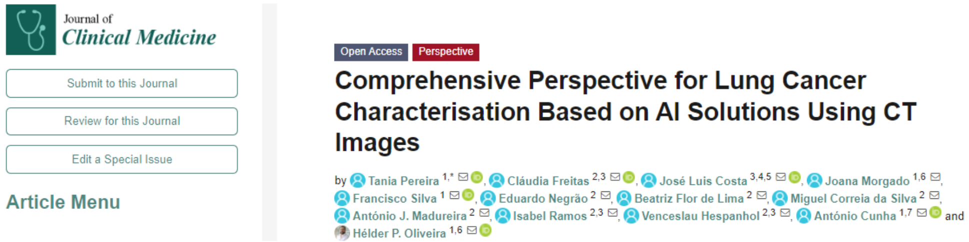
Open Access Review

Towards Machine Learning-Aided Lung Cancer Clinical Routines: Approaches and Open Challenges

by Francisco Silva [1,2,*,†], Tania Pereira [1,*,†], Inês Neves [1,3], Joana Morgado [1], Cláudia Freitas [4,5], Mafalda Malafaia [1,6], Joana Sousa [1], João Fonseca [1,6], Eduardo Negrão [4], Beatriz Flor de Lima [4], Miguel Correia da Silva [4], António J. Madureira [4,5], Isabel Ramos [4,5], José Luis Costa [5,7,8], Venceslau Hespanhol [4,5], António Cunha [1,9] and Hélder P. Oliveira [1,2]



Journal of *Clinical Medicine*

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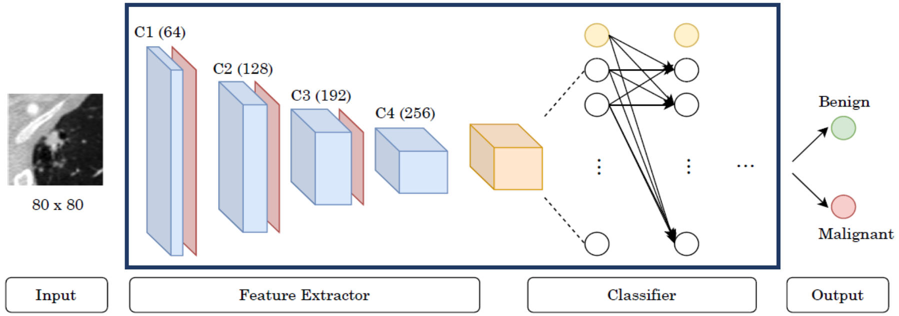
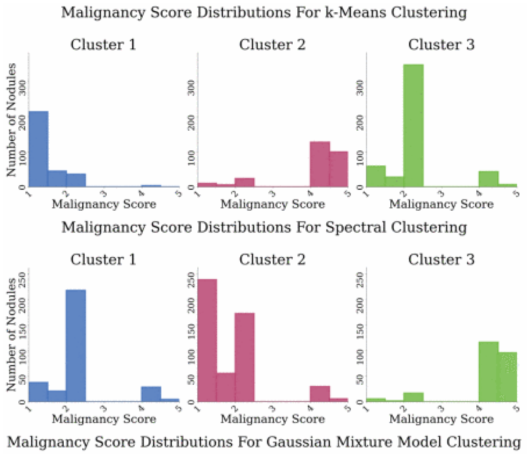
Open Access Perspective

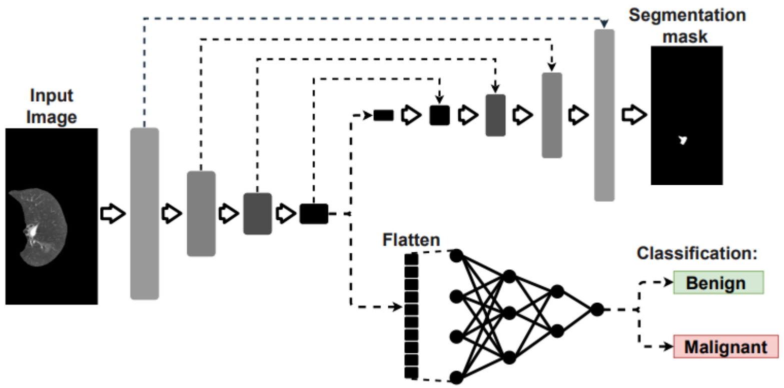
Comprehensive Perspective for Lung Cancer Characterisation Based on AI Solutions Using CT Images

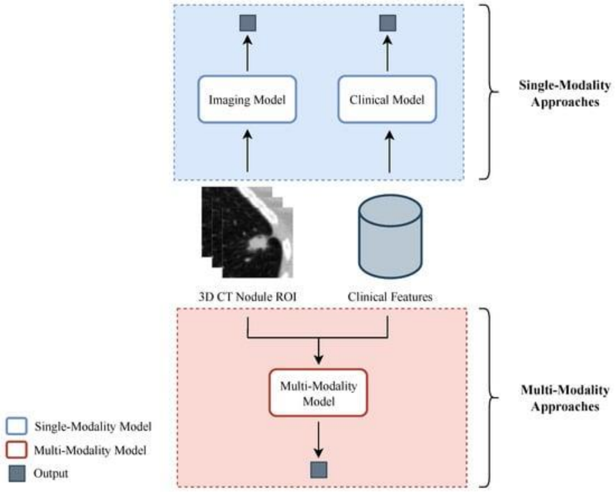
by Tania Pereira [1,*], Cláudia Freitas [2,3], José Luis Costa [3,4,5], Joana Morgado [1,6], Francisco Silva [1], Eduardo Negrão [2], Beatriz Flor de Lima [2], Miguel Correia da Silva [2], António J. Madureira [2], Isabel Ramos [2,3], Venceslau Hespanhol [2,3], António Cunha [1,7] and Hélder P. Oliveira [1,6]

MALIGNANCY PREDICTION ON CT DATA









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Single Modality vs. Multimodality: What Works Best for Lung Cancer Screening?

by Joana Vale Sousa [1,2,*], Pedro Matos [2], Francisco Silva [1,3], Pedro Freitas [1,2], Hélder P. Oliveira [1,3] and Tania Pereira [1]

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Publisher: IEEE   Cite This   PDF

Luís Fernandes ; Hélder P. Oliveira   All Authors

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Unsupervised Approach for Malignancy Assessment of Lung Nodules in Computed Tomography Scans Using Radiomic Features

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Marco Teixeira ; Tania Pereira ; Francisco Silva ; Antonio Cunha ; Helder P. Oliveira   All Authors

applied sciences

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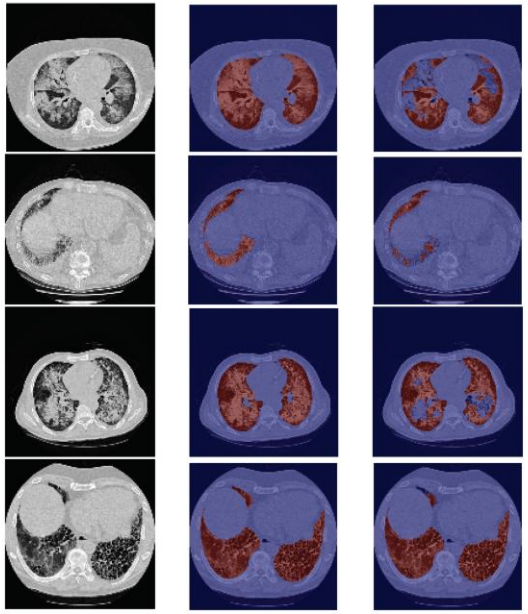
Open Access Feature Paper Article

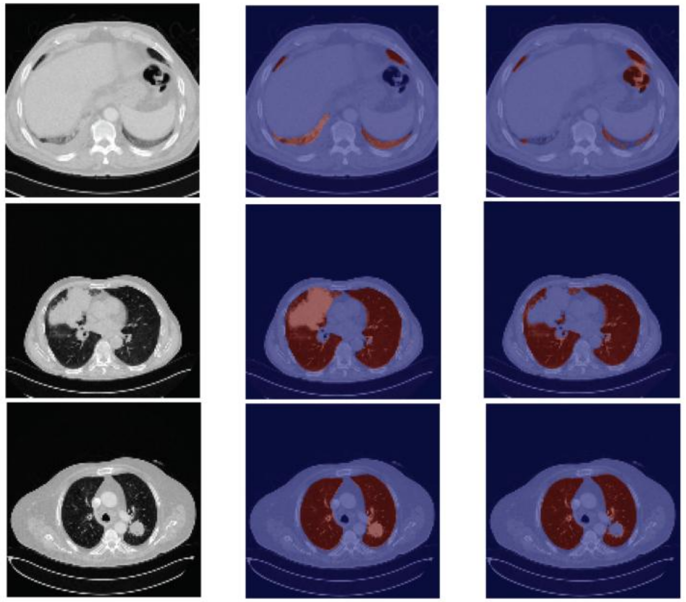
Pre-Training Autoencoder for Lung Nodule Malignancy Assessment Using CT Images

by Francisco Silva [1,2], Tania Pereira [1], Julieta Frade [1,2], José Mendes [1,2], Claudia Freitas [3,4], Venceslau Hespanhol [3,4], José Luis Costa [4,5,6], António Cunha [1,7] and Hélder P. Oliveira [1,8,*]

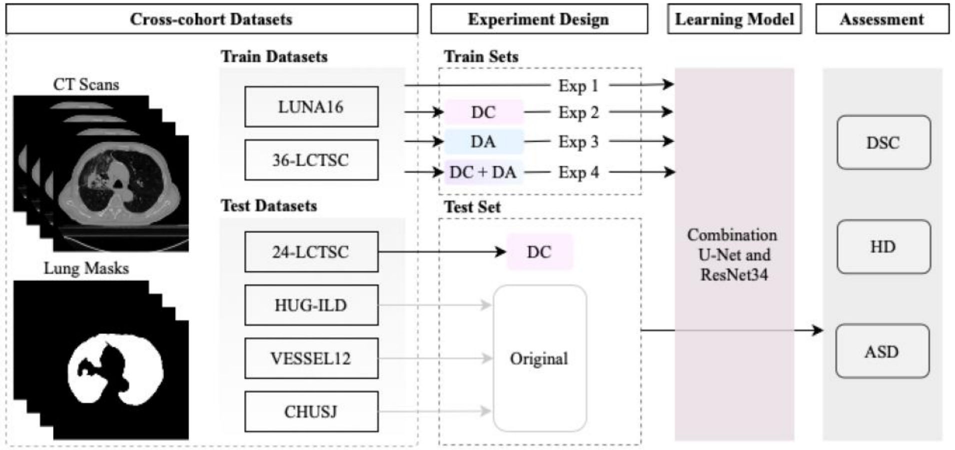
LUNG SEGMENTATION



(a) CT Image. (b) Ground Truth. (c) Predicted Mask.



(a) CT Image. (b) Ground Truth. (c) Predicted Mask.



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The Influence of a Coherent Annotation and Synthetic Addition of Lung Nodules for Lung Segmentation in CT Scans

by Joana Sousa [1,2,*], Tania Pereira [1], Inês Neves [3], Francisco Silva [1,4] and Hélder P. Oliveira [1,4]

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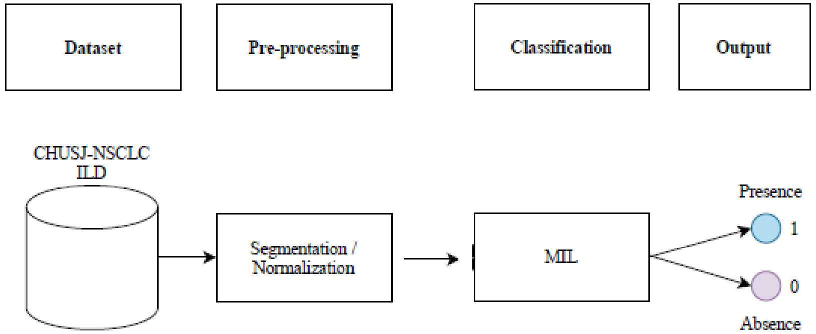
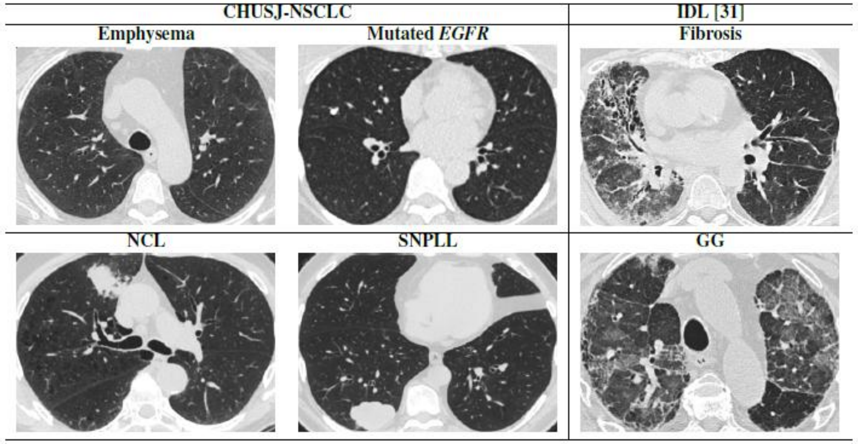
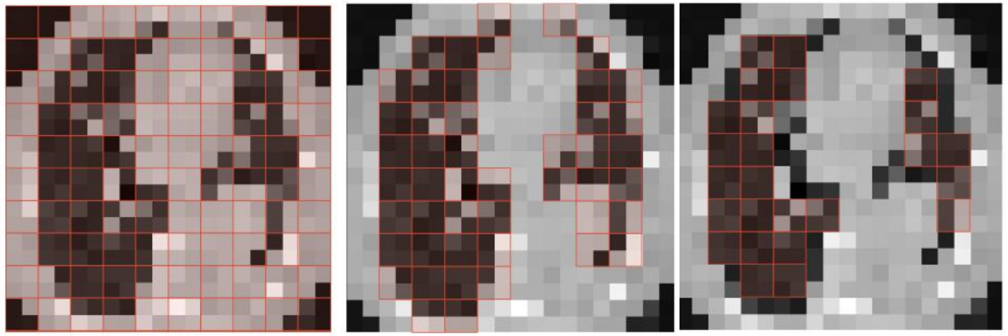
Lung Segmentation in CT Images: A Residual U-Net Approach on a Cross-Cohort Dataset

by Joana Sousa [1,2,*], Tania Pereira [1], Francisco Silva [1,3], Miguel C. Silva [4], Ana T. Vilares [4], António Cunha [1,5] and Hélder P. Oliveira [1,3]

LUNG PATHOPHYSIOLOGICAL FINDINGS







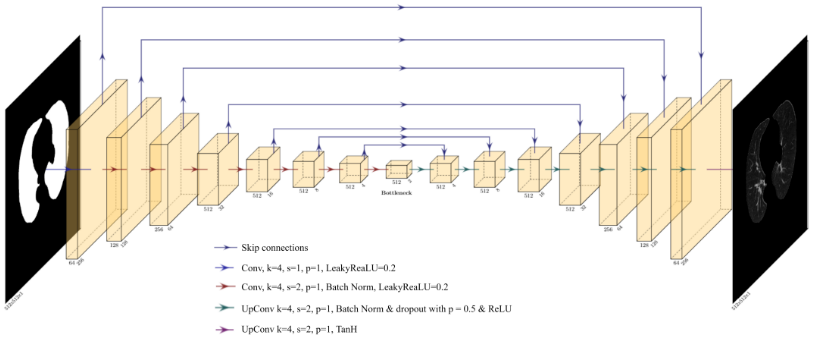
Label	AUC	Accuracy
Emphysema	0.72	0.66
EGFR	0.69	0.67
NCL	0.59	0.55
SNPLL	0.61	0.62
Fibrosis	0.89	0.83
Ground Glass	0.59	0.64

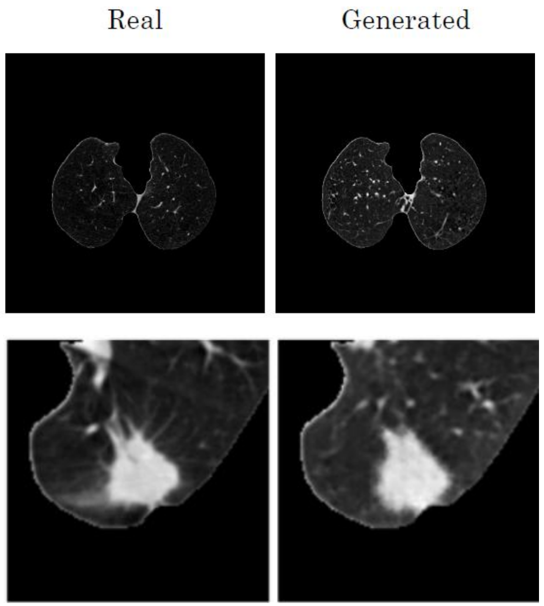
Multiple instance learning for lung pathophysiological findings detection using CT scans

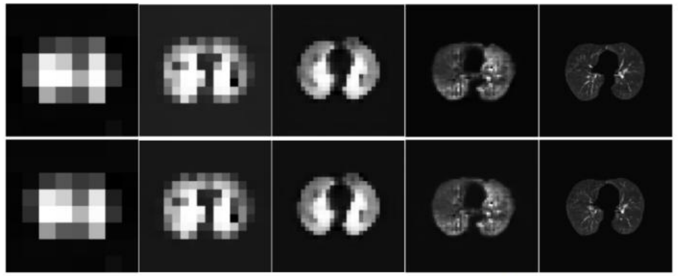
Julieta Frade, Tania Pereira, Joana Morgado, Francisco Silva, Cláudia Freitas, José Mendes, Eduardo Negrão, Beatriz Flor de Lima, Miguel Correia da Silva, António J. Madureira, Isabel Ramos, José Luís Costa, Venceslau Hespanhol, António Cunha & Hélder P. Oliveira

Medical & Biological Engineering & Computing **60**, 1569–1584 (2022) | Cite this article

CT DATA SYSNTHESIS

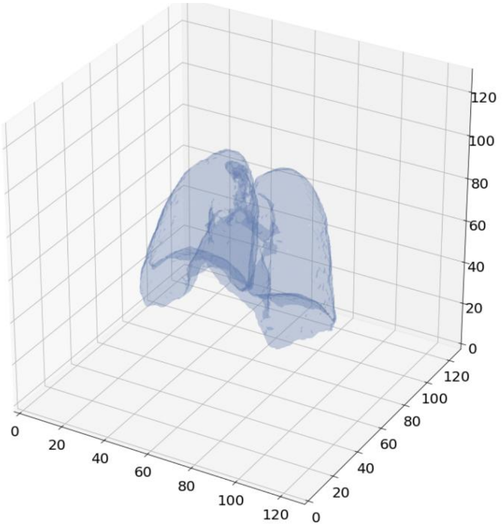






(a) 8^3 (b) 16^3 (c) 32^3 (d) 64^3 (e) 128^3

Resolution	LR G	LR D	MB	Time
4^3	0.0004	0.0005	128	5m
8^3	0.0004	0.0005	64	20m
16^3	0.0004	0.0005	32	1h 20m
32^3	0.0004	0.0005	16	6h 50m
64^3	0.0001-0.0002	0.0003-0-0005	4	2d 8h 5m
128^3	0.0002-0.0004	0.00005-0.00012	2	17d 2h 20m



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Synthesizing 3D Lung CT scans with Generative Adversarial Networks

Publisher: IEEE
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Artur Ferreira ; Tania Pereira ; Francisco Silva ; Ana T. Vilares ; Miguel C. Silva ; António Cunha ; Hélder P. Oliveira All Authors

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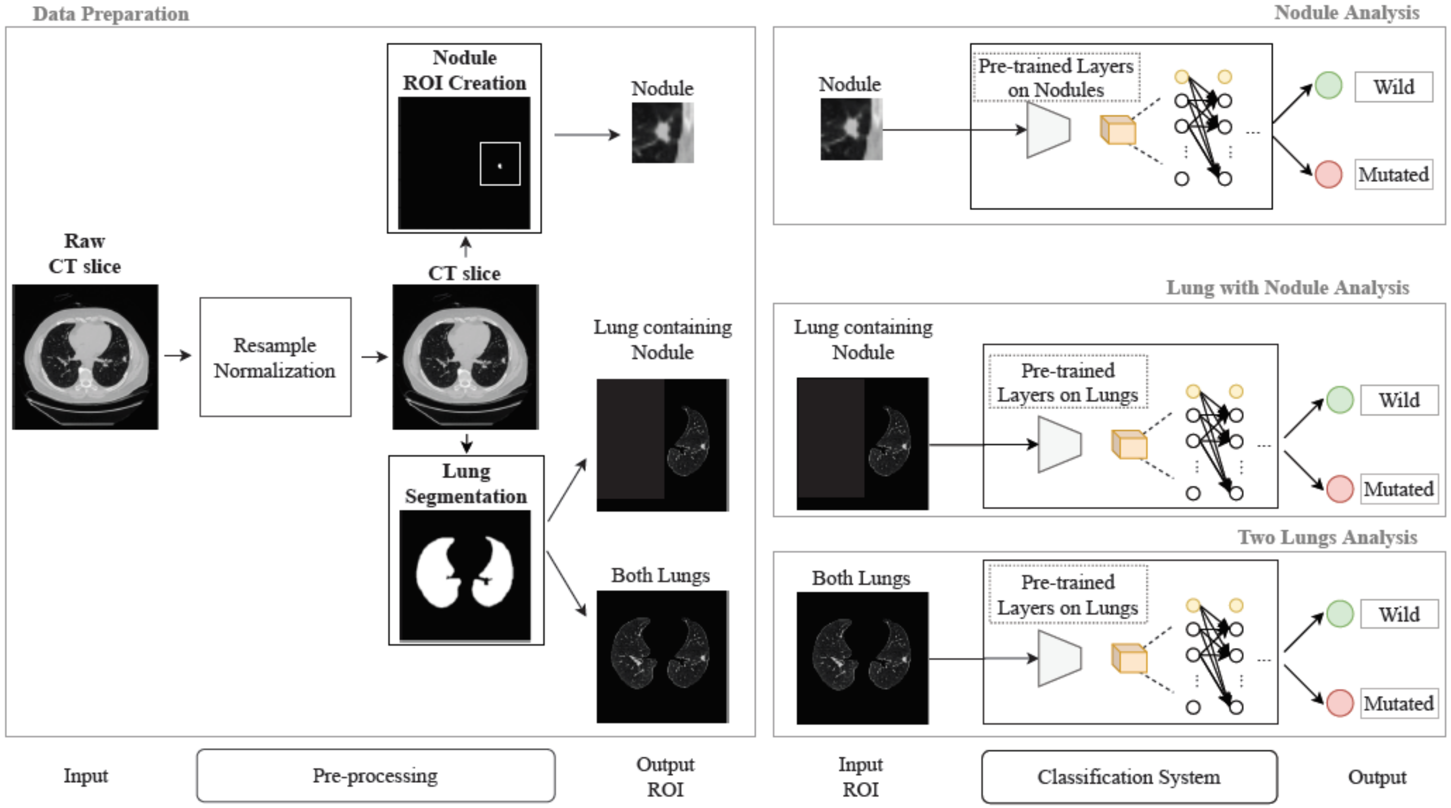
Expert Systems with Applications
Volume 215, 1 April 2023, 119350

Lung CT image synthesis using GANs

José Mendes [a b], Tania Pereira [a], Francisco Silva [a], Julieta Frade [a b], Joana Morgado [a c], Cláudia Freitas [d], Eduardo Negrão [d], Beatriz Flor de Lima [d], Miguel Correia da Silva [d], António J. Madureira [d e], Isabel Ramos [d e], José Luís Costa [e f g], Venceslau Hespanhol [d e], António Cunha [a h], Hélder P. Oliveira [a c]

EGFR MUTATION STATUS PREDICTION



Iberian Conference on Pattern Recognition and Image Analysis

↳ IbPRIA 2019: Pattern Recognition and Image Analysis pp 335–345 | Cite as

Radiogenomics: Lung Cancer-Related Genes Mutation Status Prediction

Catarina Dias, Gil Pinheiro, António Cunha & Hélder P. Oliveira

Journals & Magazines > IEEE Access > Volume: 9

EGFR Assessment in Lung Cancer CT Images: Analysis of Local and Holistic Regions of Interest Using Deep Unsupervised Transfer Learning

Publisher: IEEE Cite This PDF

Francisco Silva ; Tania Pereira ; Joana Morgado ; Julieta Frade ; José Mendes ; Cláudia Freitas ; Eduardo Negrão ; Beatriz Flor De Lima ; Mi... **All Authors**

applied sciences

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Machine Learning and Feature Selection Methods for *EGFR* Mutation Status Prediction in Lung Cancer

by Joana Morgado [1,2,*], Tania Pereira [1], Francisco Silva [1], Cláudia Freitas [3,4], Eduardo Negrão [3], Beatriz Flor de Lima [3], Miguel Correia da Silva [3], António J. Madureira [3,4], Isabel Ramos [3,4], Venceslau Hespanhol [3,4], José Luis Costa [4,5,6], António Cunha [1,7] and Hélder P. Oliveira [1,2]

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On the way for the best imaging features from CT images to predict EGFR Mutation Status in Lung Cancer

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Pedro Silva ; Tania Pereira ; Marco Teixeira ; Francisco Silva ; Hélder P. Oliveira **All Authors**

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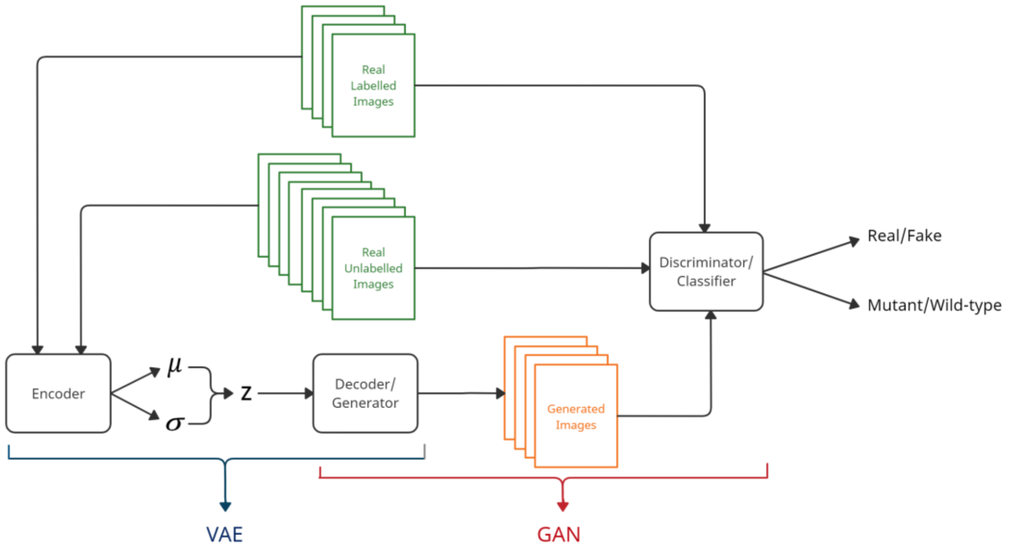
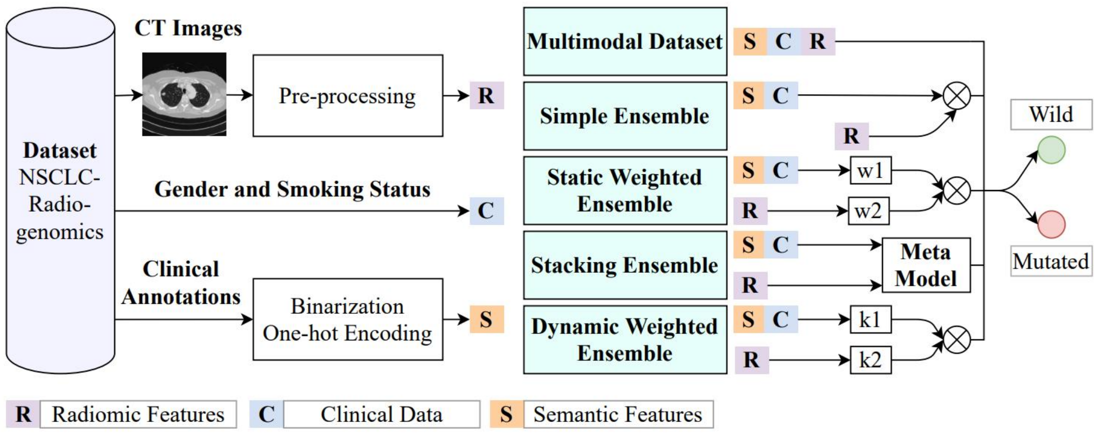
EGFR MUTATION STATUS PREDICTION





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Semi-Supervised Approach for EGFR Mutation Prediction on CT Images

by Cláudia Pinheiro [1,2], Francisco Silva [1,3], Tania Pereira [1,*] and Hélder P. Oliveira [1,3]

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Stacking Approach for Lung Cancer EGFR Mutation Status Prediction from CT Scans

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Alexandra Ventura ; Tania Pereira ; Francisco Silva ; Cláudia Freitas ; António Cunha ; Hélder P. Oliveira **All Authors**

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Ensemble Strategies for EGFR Mutation Status Prediction in Lung Cancer

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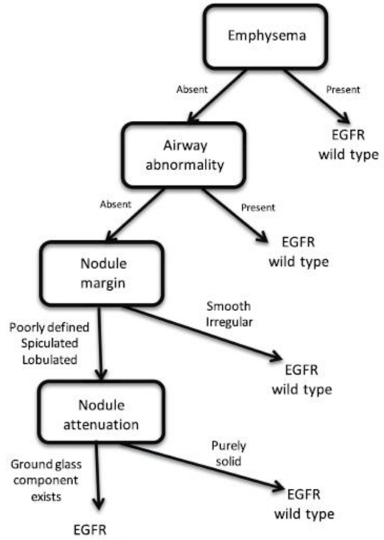
Mafalda Malafaia ; Tania Pereira ; Francisco Silva ; Joana Morgado ; António Cunha ; Hélder P. Oliveira **All Authors**

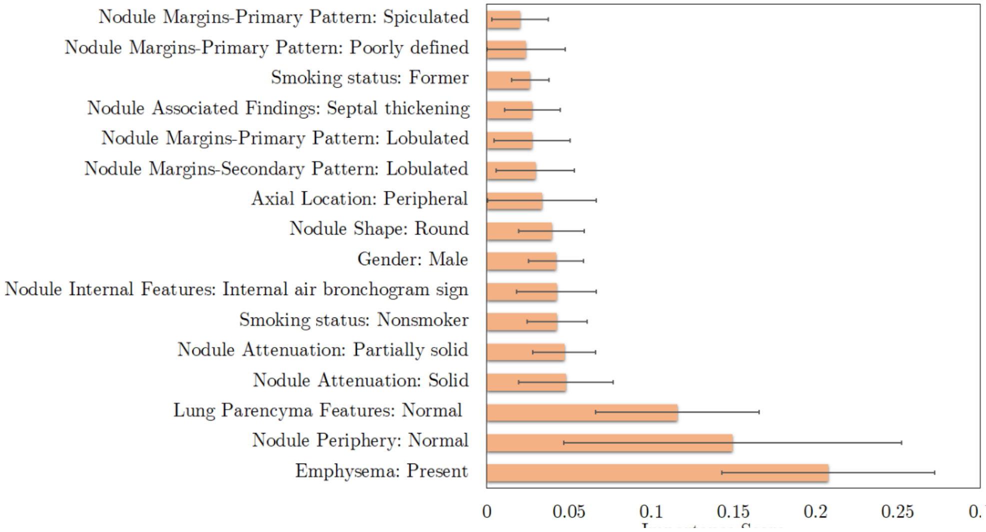
Identifying relationships between imaging phenotypes and lung cancer-related mutation status: *EGFR* and *KRAS*

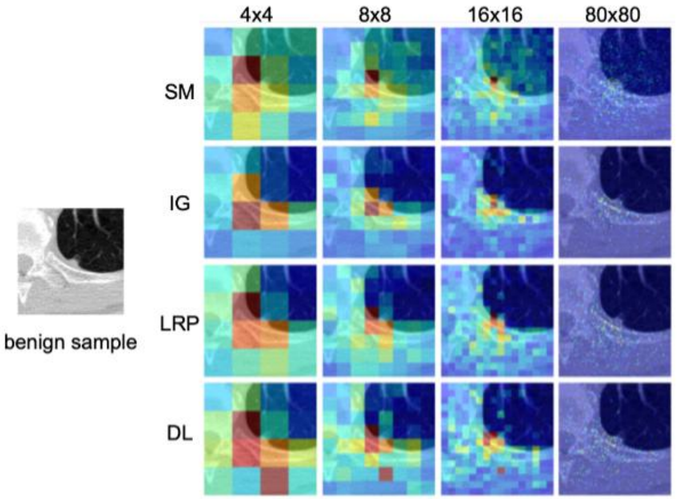
Gil Pinheiro, Tania Pereira, Catarina Dias, Cláudia Freitas, Venceslau Hespanhol, José Luis Costa, António Cunha & Hélder P. Oliveira

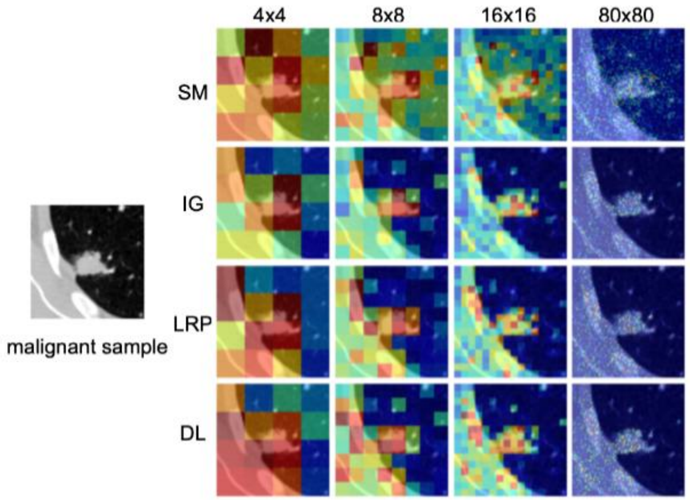
Scientific Reports **10**, Article number: 3625 (2020) | Cite this article

HOW TO HELP CLINICIANS TO UNDERSTAND THE RESULTS









Robustness Analysis of Deep Learning-Based Lung Cancer Classification Using Explainable Methods

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Mafalda Malafaia ; Francisco Silva ; Inês Neves ; Tania Pereira ; Hélder P. Oliveira All

Identifying relationships between imaging phenotypes and lung cancer-related mutation status: *EGFR* and *KRAS*

Gil Pinheiro, Tania Pereira, Catarina Dias, Cláudia Freitas, Venceslau Hespanhol, José Luis Costa, António Cunha & Hélder P. Oliveira

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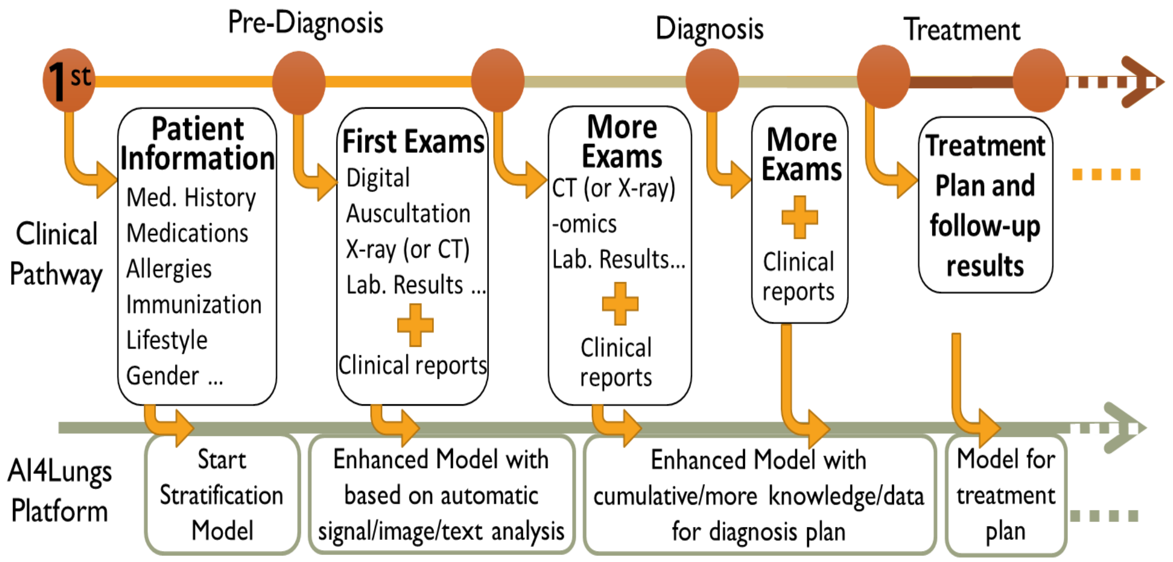
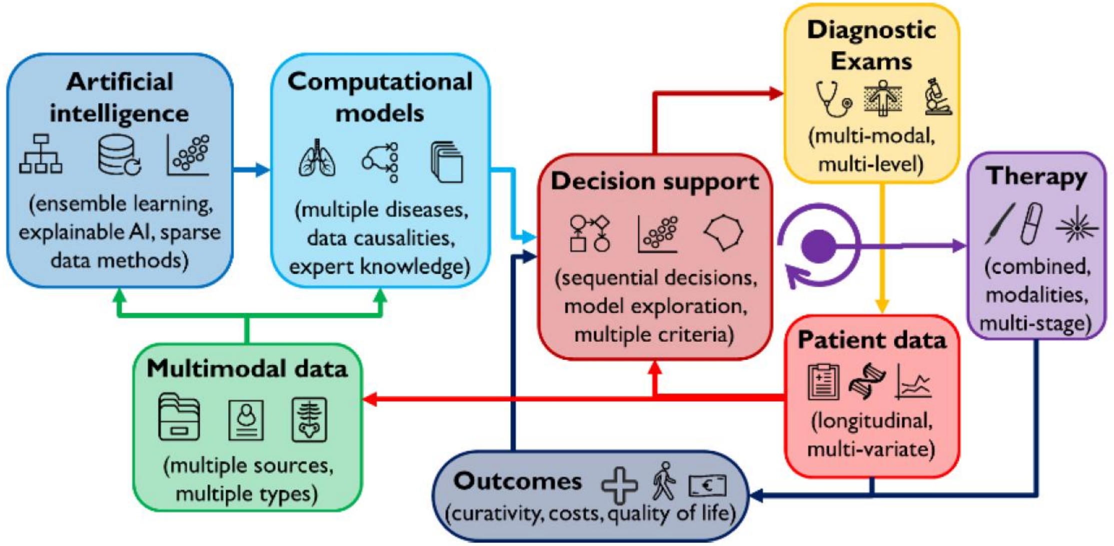
WHAT NEXT ?

Digital twin



Never ending learning



We are already doing this at AI4Lungs (European Project)

AI4LUNGS

https://www.ai4lungs.eu/

WHAT NEXT ?

AI4LUNGS

https://www.ai4lungs.eu/

- **18 Partners**
- **10 Countries**
- **6,9 Million Euros (€)**
- ***42 Months***
- ***10 Major objectives***
- ***9 Work Packages***
- ***36 Tasks***
- **37 Deliverables**
- **11 Milestones**
- **82 Members (and counting…)**

WRAP UP

WRAP UP

Cancer Research funded projects

AI4Lungs	(EU – HE)	Lung Cancer (2024 – 2027)
PHASE IV AI	(UE – HE)	Artificial Cancer Data Generation (2023 – 2026)
LUCCA	(FCT)	Lung Cancer (2023 – 2025)
CELLO	(FCT)	Blender Cancer (2023 – 2026)
CAGING	(FCT)	XAI for Cancer (2023 – 2025)
Health from Portugal	(PRR)	Lung+Breast Cancer (2022 – 2025)
CINDERELLA	(UE – HE)	Breast Cancer (2022 – 2026)
CIRCUMSTANCE	(FCT)	Microbioma (2022 – 2024)
CADPath.AI	(PT2020)	Pathology (2021 – 2022)
TAMI	(FCT - CMUPortugal)	Cervical Cancer (2020 – 2023)
LeGEM	(INESC TEC)	Breast Cancer (2019 – 2021)
LuCaS	(FCT)	Lung Cancer (2018 – 2022)
CLARE	(FCT)	Cervical Cancer (2018 – 2021)
BCCT.plan	(PT 2020)	Breast Cancer (2016 – 2019)
NanoSTIMA	(PT 2020)	Medical Imaging (2015 – 2018)
PICTURE	(EU-FP7)	Breast Cancer (2013 – 2016)
3dBCT	(FCT)	Breast Cancer (2011 – 2014)
BCCT	(FCT)	Breast Cancer (2007 – 2010)

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Computer Vision and Machine learning challenges in Cancer Research

Hélder P. Oliveira
helder.f.oliveira@inesctec.pt
holiveira@fc.up.pt

from knowledge generation to science-based innovation